

ENGLISH



ENGLISH



ENGLISH



ENGLISH

User Manual

version 1.13

TLM8



SYMBOLS

Here are the symbols used in the manual to draw the reader's attention:



Caution! Risk of electric shock.



Caution! This operation must be performed by skilled personnel.



Pay particular attention to the following instructions.



Further information.

WARRANTY

24 months from the date of the delivery note. Warranty covers only failures of defective components (due to construction defects or defects in materials) and includes replacement or repair of the components and related labor costs.

Warranty is automatically forfeited in the event of:

- tampering, deletion, removal of the identification label and/or serial number of the product
- misuse, transformation, alteration, repair of products not carried out by Laumas personnel

Laumas provides a 1-year warranty from the date of the delivery note on defects in material or manufacture of the battery.

GUIDELINES FOR PROPER DISPOSAL



**Sealed Lead Acid
Battery
Must be recycled
Properly**

This symbol on the product or packaging indicates that:

- This is electrical/electronic equipment and cannot be disposed of as municipal solid waste, but must be delivered to a recycling center
- Improper use or disposal can pollute the environment or damage human health
- Non-compliance with these guidelines will be penalized in accordance with the regulations in force in the country of destination
- It is recommended to dispose of the packing and packaging as required by local regulations

TABLE OF CONTENTS

USER WARNINGS	1
RECOMMENDATIONS FOR CORRECT INSTALLATION OF WEIGHING INSTRUMENTS...	1
RECOMMENDATIONS FOR CORRECT INSTALLATION OF THE LOAD CELLS	2
LOAD CELL INPUT TEST (QUICK ACCESS)	4
LOAD CELL TESTING	4
INTRODUCTION TO THE MANUAL	5
MAIN SPECIFICATIONS OF THE INSTRUMENT	5
VERSION IDENTIFICATION	6
TECHNICAL SPECIFICATIONS	7
KEYS AND SYMBOLS FUNCTIONS	8
MENU MAP	9
SETPOINT (BASE program)	9
SYSTEM PARAMETERS (BASE program).....	9
SYSTEM PARAMETERS (LOAD program).....	11
BATCHING CONSTANTS (LOAD program)	13
DISPLAY VISUALIZATIONS UPON INSTRUMENT SWITCH-ON.....	14
PROGRAMMING OF SYSTEM PARAMETERS	14
FILTER ON THE WEIGHT	15
<i>ANTI PEAK</i>	15
ALIBI MEMORY (BASE program).....	16
<i>ALIBI MEMORY READING</i>	16
ALIBI MEMORY (LOAD program).....	17
<i>ALIBI MEMORY READING</i>	17
OUTPUTS AND INPUTS CONFIGURATION (BASE program).....	18
OUTPUTS AND INPUTS CONFIGURATION (LOAD program)	21
ALARM RELAY CLOSURE (LOAD program).....	21
SEMI-AUTOMATIC TARE (NET/GROSS).....	22
PRESET TARE (SUBTRACTIVE TARE DEVICE)	22
SEMI-AUTOMATIC ZERO (WEIGHT ZERO-SETTING FOR SMALL VARIATIONS)	23
PEAK (BASE program)	23
ANALOG OUTPUT (TLM8 Rev 1 only)	24
SERIAL COMMUNICATION SETTING	25
<i>RS485 SERIAL COMMUNICATION</i>	27
<i>DIRECT CONNECTION BETWEEN RS485 AND RS232 WITHOUT CONVERTER</i>	27
AUTOMATIC DIAGNOSTICS OF LOAD DISTRIBUTION	28
TEST.....	29
EVENTS LOG.....	29
INFO MENU	30
SETPOINT PROGRAMMING (BASE program)	31
USE WITH W SERIES INSTRUMENTS (BASE program).....	31

CONNECTION TO THE WEIGHT INDICATOR	31
ADDITIONAL MENU MAP	32
TLM8 REMOTE CONTROL	33
<i>REMOTE KEYPAD LOCKING</i>	33
ADDITIONAL ALARMS	33
ALARMS	34
BATCHING (LOAD program)	36
INTRODUCTION TO THE OPERATION	36
BATCHING SEQUENCE	36
<i>BATCHING SEQUENCE PROGRAMMING</i>	36
<i>BATCHING START</i>	37
<i>BATCHING EXECUTION</i>	37
<i>WAITING PHASE</i>	38
<i>CYCLE END PHASE</i>	39
OPERATION SETTINGS	39
DISPLAYING DURING BATCHING	40
BATCHING STOP	40
CONSUMPTION	40
<i>CONSUMPTION DELETION</i>	40
PROGRAMMING OF BATCHING CONSTANTS	41
<i>MINIMUM WEIGHT</i>	41
<i>MAXIMUM WEIGHT</i>	41
<i>SAFE EMPTYING TIME</i>	41
<i>WAITING TIME</i>	42
<i>NO COMPARISON TIME</i>	42
<i>NO PRODUCT LOAD TIME</i>	42
<i>NO PRODUCT UNLOAD TIME</i>	42
<i>FALL</i>	43
<i>TOLERANCE</i>	44
<i>SLOW</i>	44
<i>TAPPING FUNCTION</i>	44
<i>AUTOTARE</i>	45
<i>AUTOZERO</i>	45
<i>AUTOTARE OR AUTOZERO DELAY</i>	45
<i>CONSUMPTION</i>	45
<i>PRINT AT CYCLE END</i>	46
<i>WAITING CONFIRMATION FROM PC (SLAVE)</i>	46
<i>SWITCHING OF THE ALARM RELAY ON WEIGHT</i>	46
<i>SWITCHING OF THE TOLERANCE RELAY ON WEIGHT</i>	46
FORMULA PROGRAMMING	47
<i>DELETING FORMULA</i>	47
ADDITIONAL ALARMS RELATING TO BATCHING	48
PRINTING EXAMPLES	49
INFORMATION ON TYPE-APPROVED INSTRUMENTS IN THE EUROPEAN UNION	53
DECLARATION OF CONFORMITY - EU	54
DECLARATION OF CONFORMITY -UKCA.....	55

USER WARNINGS

RECOMMENDATIONS FOR THE PROPER USE OF WEIGHING INSTRUMENT

- Keep away from heat sources and direct sunlight
- Repair the instrument from rain (except special IP versions)
- Do not wash with water jets (except special IP versions)
- Do not dip in water
- Do not spill liquid on the instrument
- Do not use solvents to clean the instrument
- Do not install in areas subject to explosion hazard (except special Atex versions)
- If the working temperature reaches the permitted limits, it is advisable to distance the instruments to ensure adequate air flow and avoid malfunctions (e.g.: sudden shutdowns or disconnections)

RECOMMENDATIONS FOR CORRECT INSTALLATION OF WEIGHING INSTRUMENTS

The terminals indicated on the instrument's wiring diagram to be connected to earth must have the same potential as the weighed structure (same earthing pit or earthing system). If you are unable to ensure this condition, connect with an earthing wire the terminals of the instrument (including the terminal –SUPPLY) to the weighed structure.

The cell cable must be individually led to its panel input and not share a conduit with other cables; connect it directly to the instrument terminal strip without breaking its route with support terminal strips. Use "RC" filters on the instrument-driven solenoid valve and remote control switch coils.

Avoid inverters in the instrument panel; if inevitable, use special filters for the inverters and separate them with sheet metal partitions.

The panel installer must provide electric protections for the instruments (fuses, door lock switch etc.). It is advisable to leave the equipment always switched on to prevent the formation of condensation.

MAXIMUM CABLE LENGTHS

- RS485: 1000 metres with AWG24, shielded and twisted cables
- Analog current output: up to 500 metres with 0.5 mm² cable
- Analog voltage output: up to 300 metres with 0.5 mm² cable

RECOMMENDATIONS FOR CORRECT INSTALLATION OF THE LOAD CELLS

SIZING OF LOAD CELLS CAPACITY

For safety reasons, in case of static weighing, it is advisable to use the load cells at a maximum of 70-80% of its nominal capacity (assuming that the load is uniformly distributed over the entire weighed structure); depending on the handling mode of the load to weigh, consider to further reduce the % of load with respect to the nominal capacity (ex.: forklifts handling, bridge cranes, etc.).

In case of weighing with dynamic loads, the installer has to estimate the thrust speed, the acceleration, the frequency, etc.

INSTALLING LOAD CELLS

The load cells must be placed on rigid, stable in-line structures; it is important to use the mounting modules for load cells to compensate for misalignment of the support surfaces.

PROTECTION OF THE CELL CABLE

Use water-proof sheaths and joints in order to protect the cables of the cells.

MECHANICAL RESTRAINTS (pipes, etc.)

When pipes are present, we recommend the use of hoses and flexible couplings with open mouthpieces with rubber protection; in case of hard pipes, place the pipe support or anchor bracket as far as possible from the weighed structure (at a distance at least 40 times the diameter of the pipe).

WELDING

Avoid welding with the load cells already installed. If this cannot be avoided, place the welder ground clamp close to the required welding point to prevent sending current through the load cell body.

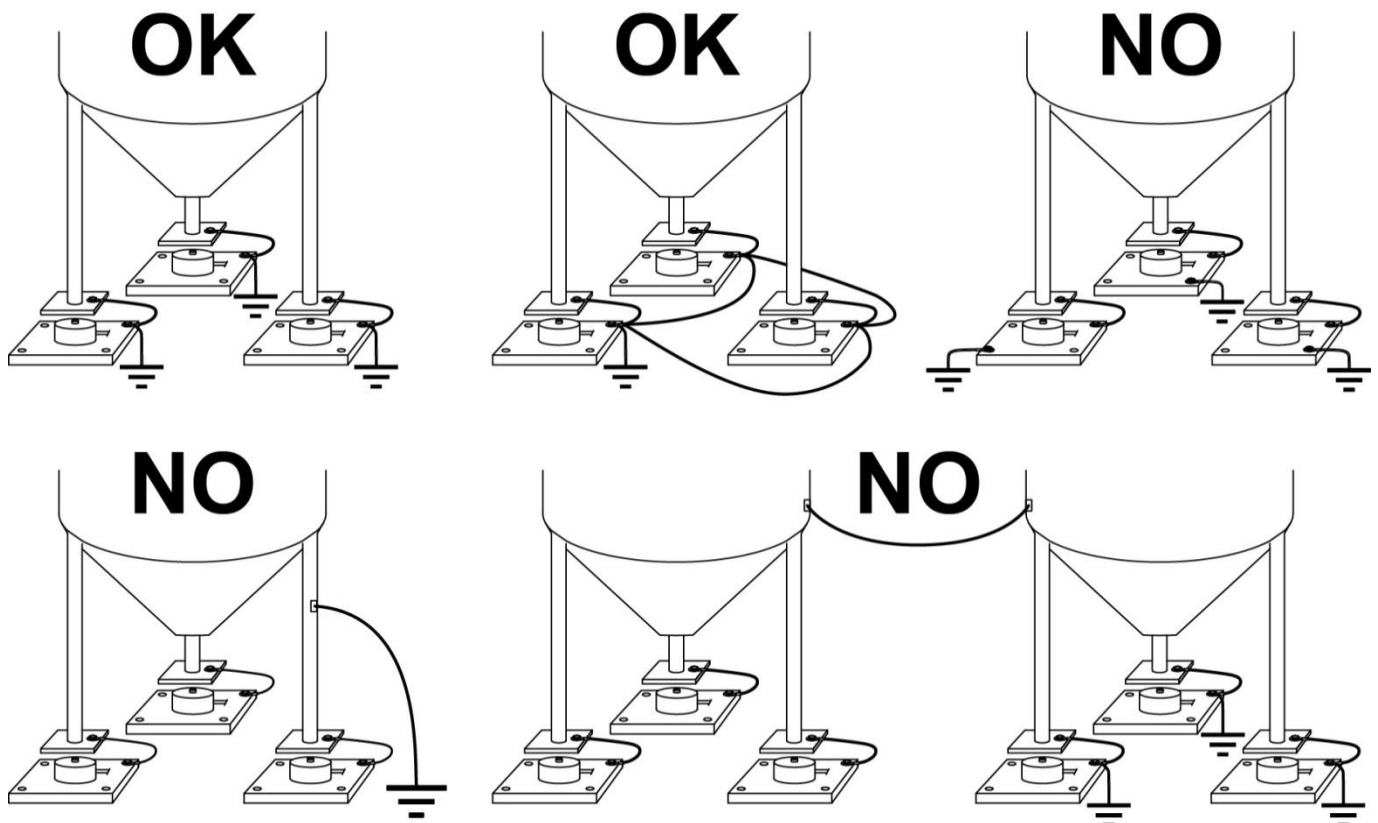
WINDY CONDITIONS - KNOCKS – VIBRATIONS

The use of weigh modules is strongly recommended for all load cells to compensate for misalignment of the support surfaces. The system designer must ensure that the plant is protected against lateral shifting and tipping relating to: shocks and vibration; windy conditions; seismic conditions in the installation setting; stability of the support structure.

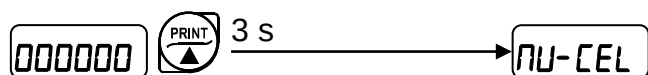
EARTHING THE WEIGHED STRUCTURE

By means of a copper wire with suitable cross-section, connect the cell upper support plate with the lower support plate, then connect all the lower plates to a single earthing system. Electrostatic charges accumulated because of the product rubbing against the pipes and the weighed container walls are discharged to the ground without going through or damaging the load cells. Failure to implement a proper earthing system might not affect the operation of the weighing system; this, however, does not rule out the possibility that the cells and connected instrument may become damaged in the future. It is forbidden to ensure earthing system continuity by using metal parts contained in the weighed structure.

FAILURE TO FOLLOW THE INSTALLATION RECOMMENDATIONS WILL BE CONSIDERED A MISUSE OF THE EQUIPMENT



LOAD CELL INPUT TEST (QUICK ACCESS)



From the weight display, press  for 3 seconds: the top of the display shows the gross weight; the bottom shows the response signal of each load cell expressed in mV with three decimals.

Example: a load cell with 2.000 mV/V sensitivity provides a response signal between 0 and 10 mV.

5130		mV	
1 OFF	2 6.468	3 OFF	4 3.480
5 OFF	6 OFF	7 OFF	8 5.441

LOAD CELL TESTING

Load cell resistance measurement (use a digital multimeter):

- Turn off the instrument.
- The value between the positive signal wire and the negative signal wire must be equal or similar to the one indicated in the load cell data sheet (output resistance).
- The value between the positive excitation wire and the negative excitation wire must be equal or similar to the one indicated in the load cell data sheet (input resistance).

Load cell voltage measurement (use a digital multimeter):

- Turn on the instrument.
- Take out the load cell to be tested from underneath the container, or alternatively, lift the container support.
- Make sure that the excitation of two wires of the load cell connected to the instrument (or amplifier) is 5 VDC $\pm 3\%$.
- Measure the response signal between the positive and the negative signal wires by directly connecting them to the tester, and make sure that it is comprised between 0 and ± 0.5 mV.
- Apply load to the cell and make sure that there is a signal increment.

IF ONE OF THE ABOVE CONDITIONS IS NOT MET, PLEASE CONTACT THE TECHNICAL ASSISTANCE SERVICE.

INTRODUCTION TO THE MANUAL

The instrument is available in two versions and the manual is valid for both:

- TLM8 Rev 1: with analog output
- TLM8 Rev 2: without analog output

The specific characteristics of only one of the two versions will be suitably indicated.

MAIN SPECIFICATIONS OF THE INSTRUMENT

Weight transmitter with 6-wire load cells inputs suitable for assembly on back panel fitted Omega/DIN rail. Dimensions: 148x92x60 mm. Backlit LCD graphic display, 128x64 pixel resolution, 60x32 mm viewing area. 5-key keypad.

RS485 serial port for connection to: PC/PLC up to 32 instruments (max 99 with line repeaters) by ASCII Laumas or ModBus R.T.U. protocol, remote display, printer. Optional: integrated Profibus DP, DeviceNet, CANopen, Profinet IO, Ethernet/IP, Ethernet TCP/IP, Modbus TCP, RS232/RS485, SERCOS III, PowerLink, CC-Link output.

TLM8 Rev 1 only: 16 bit analog output (tension or current).

8 independent channels: automatic detection of connected load cells.

Digital equalization: load cell response uniformed via software.

Load distribution: graph showing the weight percentage on each load cell.

Automatic diagnostics: load distribution check to detect any faults.

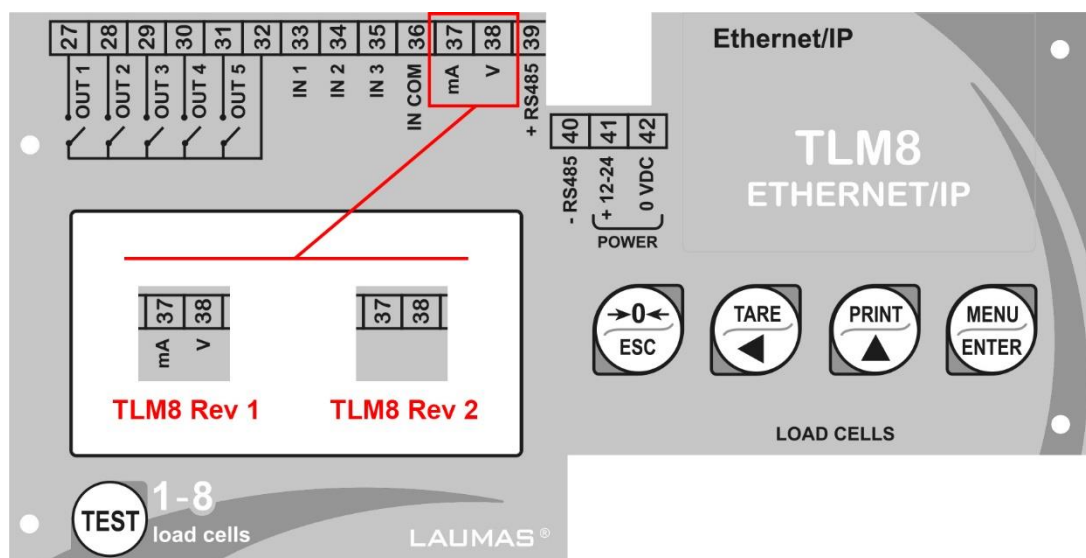
Overload diagnostics: overload check on each load cell.

Events log: storage of the last 50 events: calibrations, zero-settings, errors, equalizations.

Alarm Relay: outputs can be set to switch in case of alarm.

VERSION IDENTIFICATION

To define the version of the instrument, check the presence on the cover of the indications relating to the analog output.



TECHNICAL SPECIFICATIONS

POWER SUPPLY and CONSUMPTION	12/24 VDC $\pm 10\%$; 5 W
NO. OF LOAD CELLS IN PARALLEL and SUPPLY	max 16 (350 ohm); 5 VDC/240 mA
LINEARITY	$< 0.01\%$ F.S.
ANALOG OUTPUT LINEARITY (TLM8 Rev 1 only)	$< 0.01\%$ F.S.
THERMAL DRIFT	$< 0.0005\%$ F.S./ $^{\circ}\text{C}$
ANALOG OUTPUT THERMAL DRIFT (TLM8 Rev 1 only)	$< 0.003\%$ F.S./ $^{\circ}\text{C}$
A/D CONVERTER	8 channels, 24 bit (16000000 points), 4.8 kHz
DIVISIONS (with measurement range ± 10 mV = sens. 2 mV/V)	± 999999
MEASUREMENT RANGE	± 39 mV
MAX SENSITIVITY OF USABLE LOAD CELLS	± 7 mV/V
MAX CONVERSIONS PER SECOND	600 conversions/second
DISPLAY RANGE	± 999999
NO. OF DECIMALS / DISPLAY INCREMENTS	0÷4 / x 1 x 2 x 5 x 10 x 20 x 50 x 100
DIGITAL FILTER / READINGS PER SECOND	21 levels / 5÷600 Hz
RELAY OUTPUTS	N. 5 - max 115 VAC; 150 mA
DIGITAL INPUTS	N. 3 - optoisolated 5 - 24 VDC PNP
SERIAL PORTS	RS485
BAUD RATE	1200, 2400, 4800, 9600, 19200, 38400, 115200
HUMIDITY (non condensing)	85%
STORAGE TEMPERATURE	-30 $^{\circ}\text{C}$ +80 $^{\circ}\text{C}$
WORKING TEMPERATURE	-20 $^{\circ}\text{C}$ +60 $^{\circ}\text{C}$
ANALOG OUTPUT (TLM8 Rev 1 only) 16 bit - 65535 divisions	0÷20 mA; 4÷20 mA (max 400 ohm); 0÷10 V; 0÷5 V (min 2 kohm)

	RELAY OUTPUTS	N. 5 - max 30 VAC, 60 VDC; 150 mA
	WORKING TEMPERATURE	-20 $^{\circ}\text{C}$ +60 $^{\circ}\text{C}$
	Equipment to be powered by 12-24 VDC LPS or Class 2 power source.	

METROLOGICAL SPECIFICATIONS OF TYPE-APPROVED INSTRUMENTS	OIML	INMETRO
APPLIED STANDARDS BY REGION	EU: 2014/31/EU; OIML R76:2006; EN45501:2015 United Kingdom: Non-automatic Weighing Instrument Regulations 2016	Brazil: Portaria Inmetro N°157/2022
OPERATION MODE	single interval, multi-interval, multiple range	single interval, multi-interval, multiple range
ACCURACY CLASS	III or IIII	III
MAX NUMBER OF SCALE VERIFICATION DIVISIONS	10000 (class III); 1000 (class IIII)	10000 (class III)
MINIMUM INPUT SIGNAL FOR SCALE VERIFICATION DIVISION	0.2 μV	0.2 μV
WORKING TEMPERATURE	-10 $^{\circ}\text{C}$ +40 $^{\circ}\text{C}$	-10 $^{\circ}\text{C}$ +40 $^{\circ}\text{C}$

KEYS AND SYMBOLS FUNCTIONS

KEY	Short press	Long press (3 s)	Into menus
	Semi-automatic zero		Cancel or return to previous menu
	Gross → Net	Net → Gross	Select figure to be modified or go to previous menu item.
	Weight print	Gross weight: mV load cell test Net weight: temporarily display the gross weight	Modify selected figure or go to next menu item
	Setting setpoint and hysteresis		Confirm or enter in submenu
	Setting general parameters (press  immediately followed by )		
	Setting preset tare (press  immediately followed by )		
	Load distribution display		

SYMBOL	Function
	net weight (semi-automatic tare or preset tare)
	zero (deviation from zero not more than ±0.25 divisions)
	stability
	unit of measure: kg
	unit of measure: g
	current range in multiple range or multi-interval instruments



The symbols are activated in sequence within the menus to indicate that the display is not showing a weight.

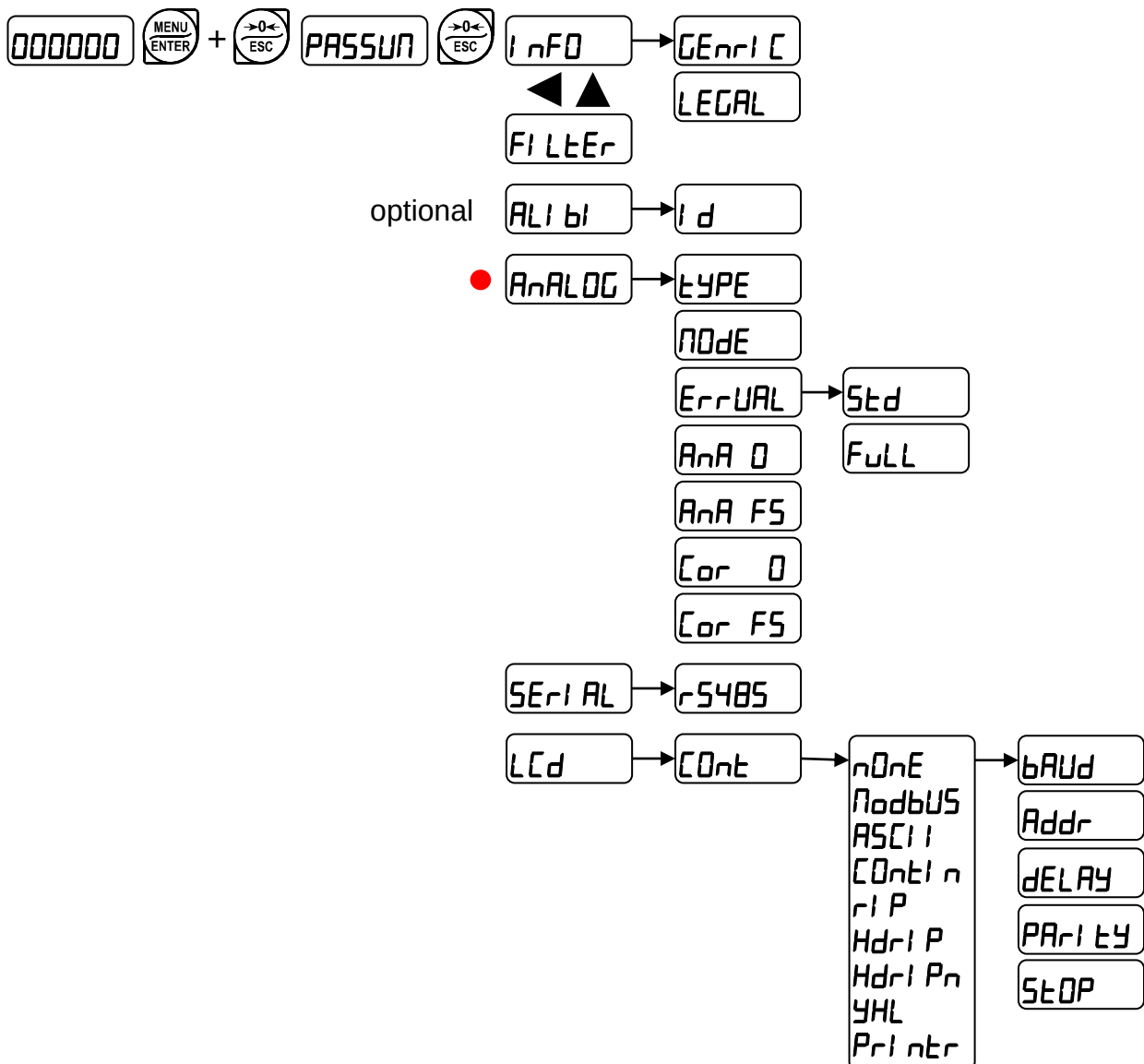
MENU MAP

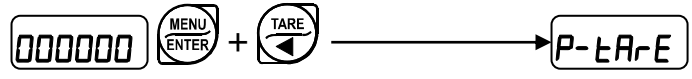
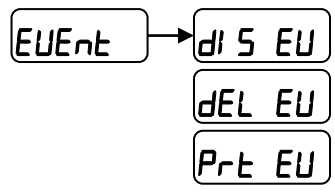
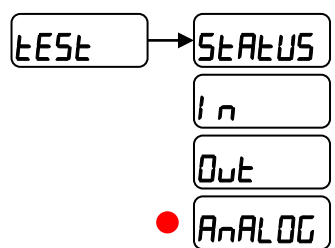
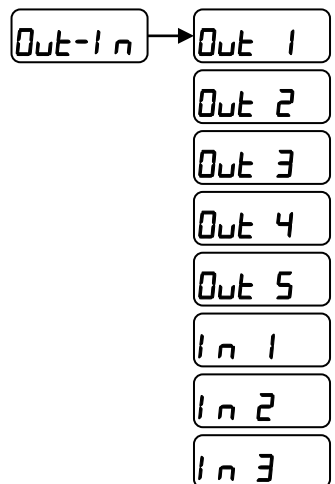
Into menus changes are applied right after pressing the **ENTER** key (no further confirmation is required).

SETPOINT (BASE program)



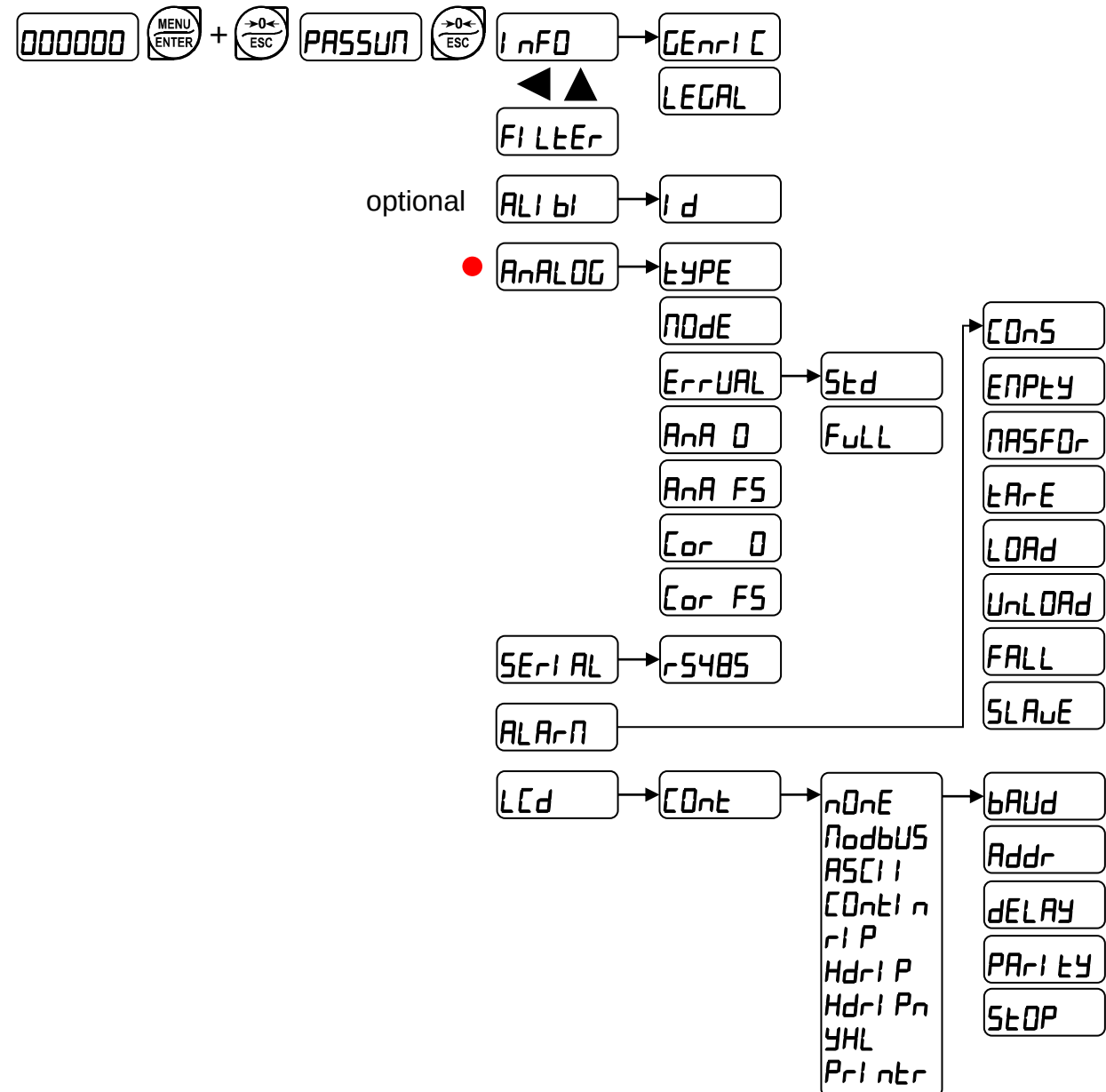
SYSTEM PARAMETERS (BASE program)

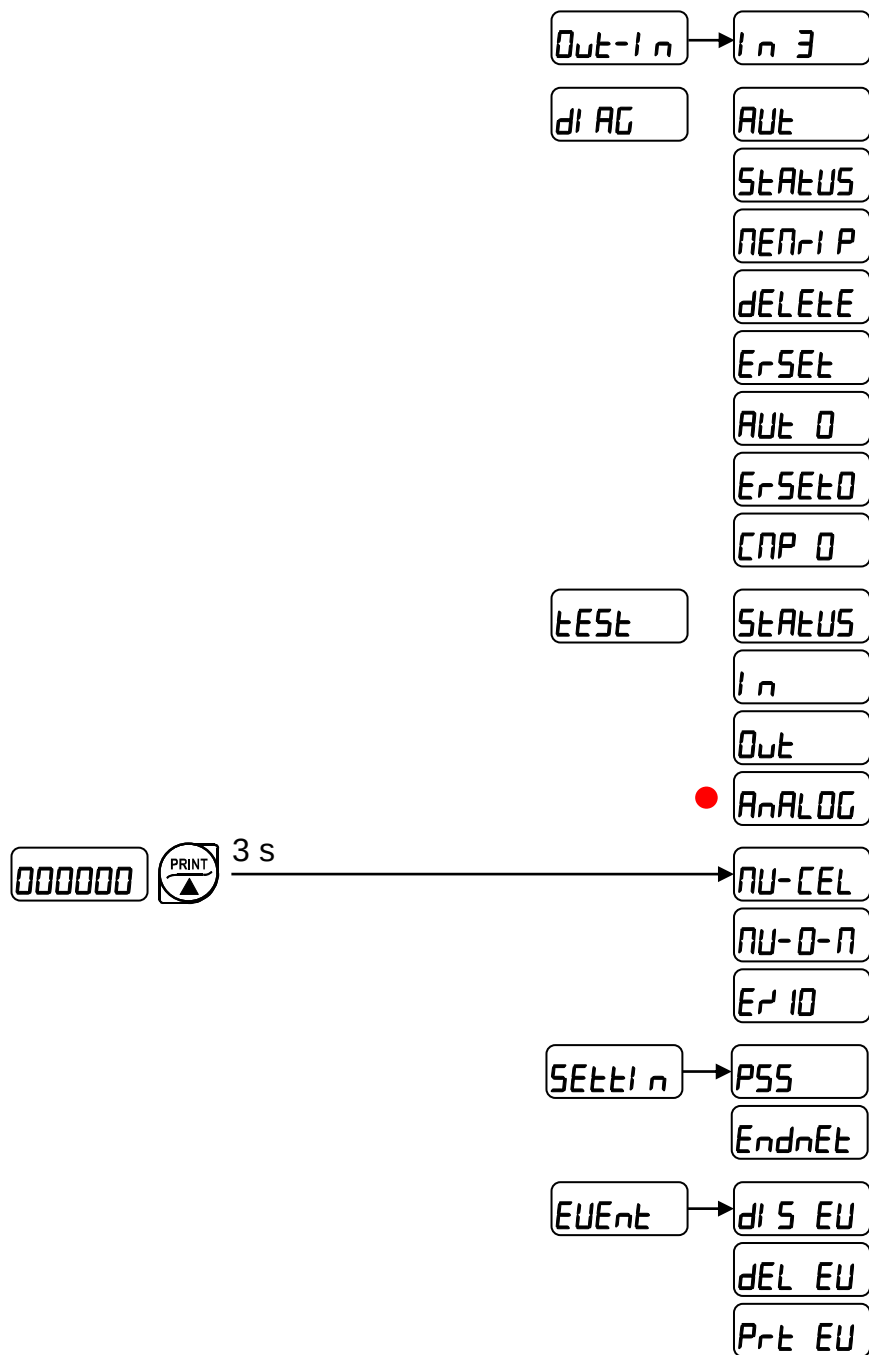




● TLM8 Rev 1 only

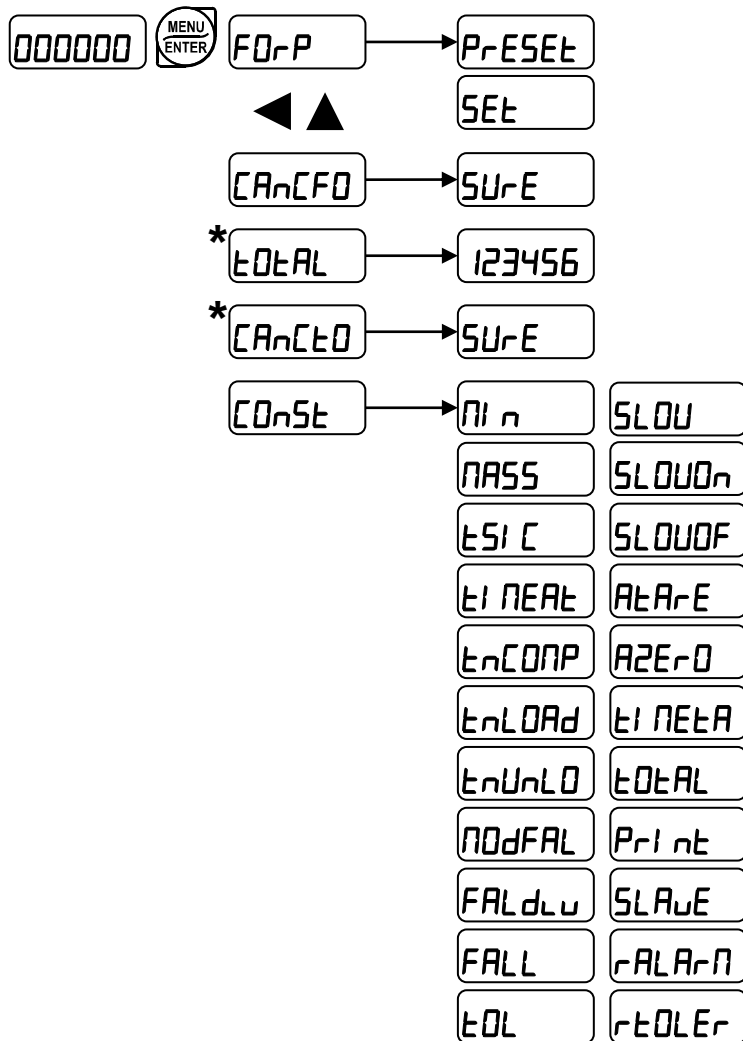
SYSTEM PARAMETERS (LOAD program)	
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	1
10	1
11	1
12	1
13	1
14	1
15	1
16	1
17	1
18	1
19	1
20	1
21	1
22	1
23	1
24	1
25	1
26	1
27	1
28	1
29	1
30	1
31	1
32	1
33	1
34	1
35	1
36	1
37	1
38	1
39	1
40	1
41	1
42	1
43	1
44	1
45	1
46	1
47	1
48	1
49	1
50	1
51	1
52	1
53	1
54	1
55	1
56	1
57	1
58	1
59	1
60	1
61	1
62	1
63	1
64	1
65	1
66	1
67	1
68	1
69	1
70	1
71	1
72	1
73	1
74	1
75	1
76	1
77	1
78	1
79	1
80	1
81	1
82	1
83	1
84	1
85	1
86	1
87	1
88	1
89	1
90	1
91	1
92	1
93	1
94	1
95	1
96	1
97	1
98	1
99	1
100	1





● TLM8 Rev 1 only

BATCHING CONSTANTS (LOAD program)



*: it appears only if in constants TOTAL = YES

DISPLAY VISUALIZATIONS UPON INSTRUMENT SWITCH-ON

Upon switch-on, the display shows in sequence:

- *111111* → *999999* (ONLY in case of approved program);
- instrument model (e.g.: *EL78*);
- *SU* followed by the software code (e.g.: *SU 100*);
- program type: *BASE* (base);
- *r* followed by the software version (e.g.: *r 1.04.01*);
- *HU* followed by the hardware code (e.g.: *HU 600*);
- serial number (e.g.: *1005 15*);

PROGRAMMING OF SYSTEM PARAMETERS

From the weight display, press simultaneously keys **MENU** and **ESC** to access the parameter setting.

MENU/ENTER: to enter a menu/confirm the data entry.



: to modify the displayed figure or menu item.



: to select a new figure or modify the displayed menu item.

ESC:

to cancel and return to the previous menu.

FILTER ON THE WEIGHT



Setting this parameter allows a stable weight display to be obtained.

Two filter types are available:

- **TYPE 0**: Laumas standard filter (default);
- **TYPE 1**: moving average filter with fixed refresh frequency.

To increase the effect (weight more stable) increase the filter level (from 0 to 9, default 4).

Procedure:

- By confirming the **Filter** message, the currently programmed filter type is displayed.
- Select the desired filter type and confirm, the currently programmed filter level is displayed.
- Select the desired filter level and confirm, the weight is displayed and it will be possible to experimentally verify its stability.
- If stability is not satisfactory, repeat the procedure until an optimum result is achieved.

The filter enables to stabilise a weight as long as its variations are smaller than the corresponding “response time”. It is necessary to set this filter according to the type of application and to the full scale value set.

LEVEL	Response time* [ms]		Display and serial port refresh frequency* [Hz]		
	TYPE 0	TYPE 1	TYPE 0	TYPE 1	
				1-4 active channels	5-8 active channels
0	12	100	300	100	50
1	150	330	100	100	50
2	260	500	50	100	50
3	425	700	25	100	50
4	850	1100	12.5	100	50
5	1700	1600	12.5	100	50
6	2500	2700	12.5	100	50
7	4000	3500	10	100	50
8	6000	5000	10	100	50
9	7000	6800	5	100	50
A	6	—	600	—	

*indicative values

The "A" filter can only be set if the instrument is connected to one load cell only.

ANTI PEAK

When the weight is stable, the anti peak filter removes any sudden disturbances with a maximum duration of 1 second. Confirm the filter on the weight with **ENTER** and select one of the following options:

- **AntPeOn**: anti peak filter enabled (default);
- **AntPeOff**: anti peak filter disabled.

ALIBI MEMORY (BASE program)



To save a weight to the alibi memory press the key for less than 3 seconds or close the dedicated input or send the command via serial protocol (see the Protocols manual).

If the net function is active, the net weight is stored; otherwise, the gross weight is stored.

Saving to the alibi memory is only possible if:

- the alibi memory is enabled;
- the weight is stable;
- the weight is above the minimum weight (20 e);
- the weight is below the max capacity;
- the weight has changed since the latest storage to alibi.

An identifying number is automatically assigned to each record stored in the memory, it increases for stored records from 1 to 999999, and is automatically reset to 1 after reaching 999999.

WARNING: the alibi memory cannot store all the 999999 records; the number of records that can be stored depends on the instrument type and on the program used.

The alibi memory is used in a “circular” manner: when the memory end is reached, the system will restart from the memory beginning by overwriting the first record; an alarm message is displayed (*ALI FUL*).

ALIBI MEMORY READING

I d (identification): enter the ID number of your required record and confirm, the display will show:

- Gross weight (*GROSS*) or net weight (*NET*); confirm with
- Tara weight (*TARE*) or preset tare (*PTARE*), only if the recalled weight is a net weight; confirm with

The parameter provides the ID of the last record stored in Alibi Memory by default.

If the entered identifier does not match any record, the message *Error* is displayed.

ALIBI MEMORY (LOAD program)



Each batched weight is saved to the alibi memory with a specific ID.

Saving to the alibi memory is only possible if:

- the alibi memory is enabled;
- the weight is stable;
- the weight is above the minimum weight (20 e);
- the weight is below the max capacity;

An identifying number is automatically assigned to each record stored in the memory, it increases for stored records from 1 to 999999, and is automatically reset to 1 after reaching 999999.

WARNING: the alibi memory cannot store all the 999999 records; the number of records that can be stored depends on the instrument type and on the program used.

The alibi memory is used in a “circular” manner: when the memory end is reached, the system will restart from the memory beginning by overwriting the first record; an alarm message is displayed (*ALI FUL*) upon batching start when more than 90% of the memory is already full.

ALIBI MEMORY READING

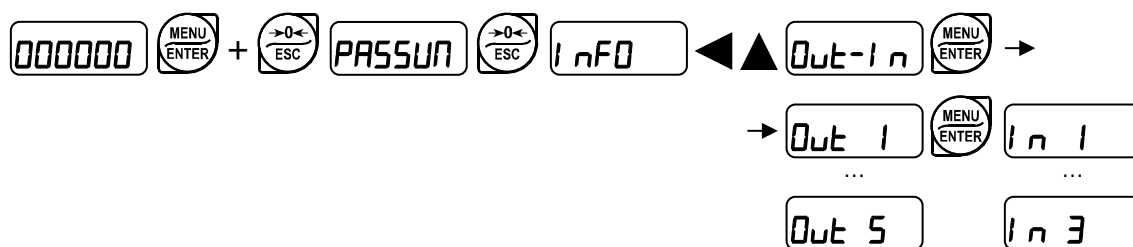
I d (identification): enter the ID number of your required record and confirm, the display will show:

- Gross weight (*GROSS*) or net weight (*NET*); confirm with **ENTER**;
- Tara weight (*TARE*) or preset tare (*PTARE*), only if the recalled weight is a net weight; confirm with **ENTER**;

The parameter provides the ID of the last record stored in Alibi Memory by default.

If the entered identifier does not match any record, the message *Error* is displayed.

OUTPUTS AND INPUTS CONFIGURATION (BASE program)



OUTPUTS

The outputs are set by default as follows: *OPEN / SET / GROSS / POSNEG / OFF*.

Possible operation modes:

- **OPEN (normally open)**: the relay is de-energised and the contact is open when the weight is lower than the programmed setpoint value; it closes when the weight is higher than or equal to the programmed setpoint value.
- **CLOSE (normally closed)**: the relay is energised and the contact is closed when the weight is lower than the programmed setpoint value; it opens when the weight is higher than or equal to the programmed setpoint value.
- **SET**: the contact will switch on the basis of weight, according to setpoint (see section **SETPOINT PROGRAMMING (BASE program)**).
- **PLC**: the contact will not switch on the basis of weight, but is controlled by remote protocol commands.
- **STABLE**: relay switching occurs when the weight is stable.
- **ALARM**: relay switching occurs when one of the following alarms is triggered: *ErCEL*, *ErCELR*, *ErCEL I*, *Er OL*, *Er Ad*, "-----", *Er OF*; the operation mode is forced to **CLOSE** (normally closed).
- **CHDL**: relay switching occurs when one of the following conditions occurs:
 - the gross weight value on the single channel is higher than the value of parameter *CHSET*;
 - the gross weight value detected by the system is higher than the value specified in the setpoints.

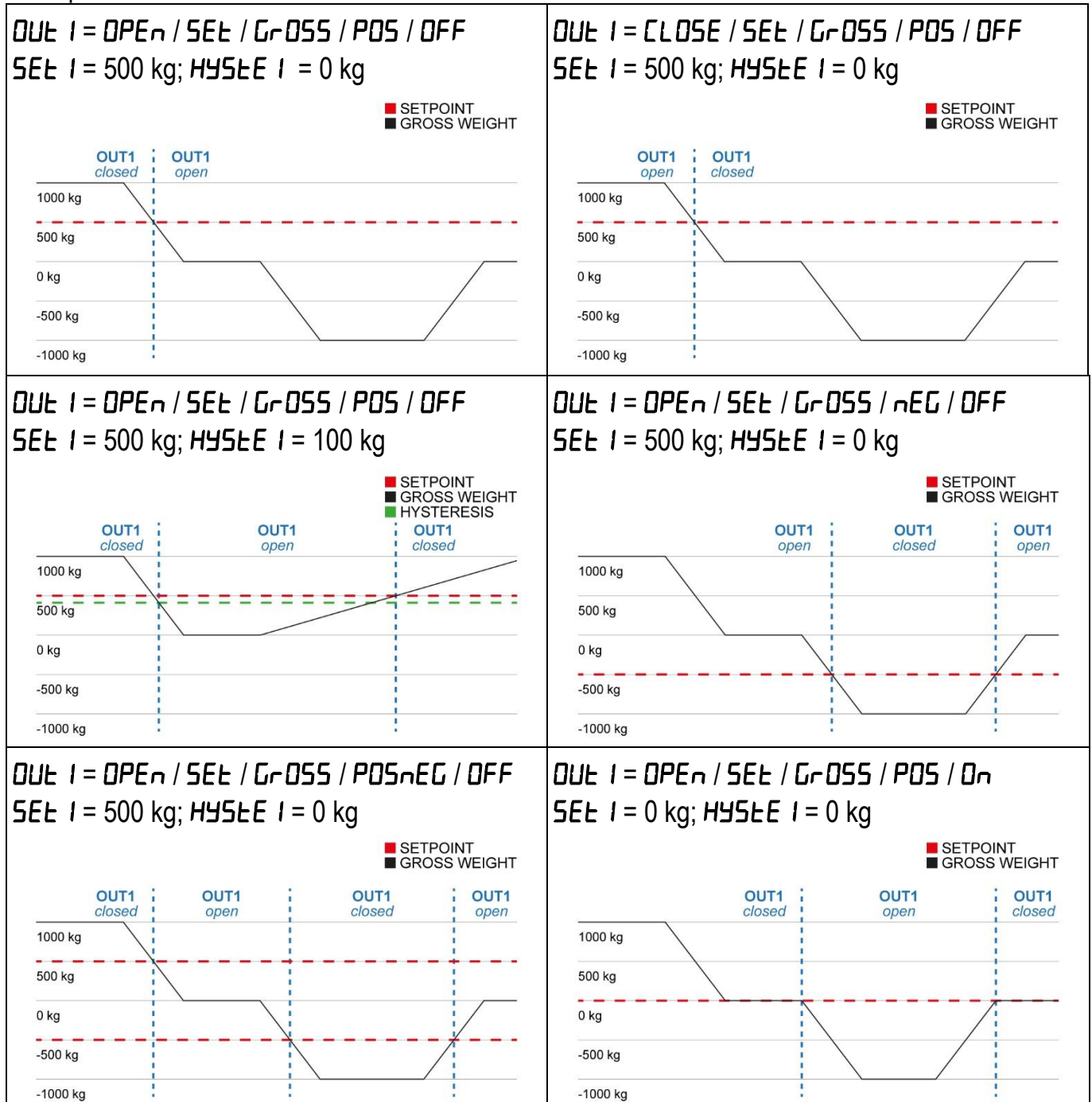
If the operation mode **SET** is selected, the following options are also active:

- **GROSS**: the contact will switch on the basis of gross weight.
- **NET**: the contact will switch on the basis of net weight (If the net function is not active, the contact will switch on the basis of gross weight).
- **POSNEG**: relay switching occurs for both positive and negative weight values.
- **POS**: relay switching occurs for positive weight values only.
- **NEG**: relay switching occurs for negative weight values only.

By confirming with **ENTER** the setpoint operation can be set to the value 0:

- **OFF**: relay switching will not occur if the setpoint value is 0.
- **On**:
 - setpoint = 0 and switching = **POSnEG**: relay switching occurs when the weight is 0; the relay will switch again when the weight is different from zero, taking hysteresis into account (both for positive and for negative weights).
 - setpoint = 0 and switching = **POS**: relay switching occurs for a weight higher than or equal to 0, the relay will switch again for values below 0, taking hysteresis into account.
 - setpoint = 0 and switching = **nEG**: relay switching occurs for a weight lower than or equal to 0, the relay will switch again for values above 0, taking hysteresis into account.

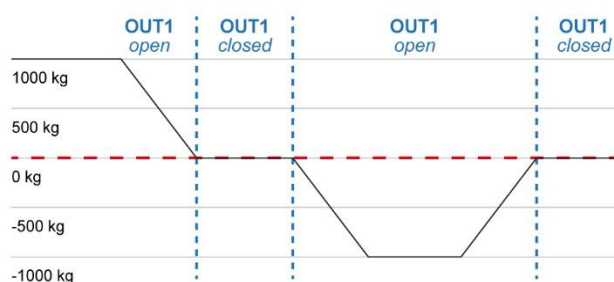
Examples:



OUT1 = OPEN / SET / CROSS / POSNEG / ON

SET 1 = 0 kg; HYSIE 1 = 0 kg

■ SETPOINT
■ GROSS WEIGHT



INPUTS

Default: input 1 = ZERO input 2 = NET-GROSS input 3 = PEAK

Possible operation modes:

- **NET-GROSS** (NET/GROSS): by closing this input for no more than one second, it's making an operation of SEMI-AUTOMATIC TARE and the display will show the net weight. To display the gross weight again, hold the NET/GROSS input closed for 3 seconds.
- **ZERO**: by closing the input for no more than one second, the weight is set to zero (see section **WEIGHT ZERO-SETTING FOR SMALL VARIATIONS (SEMI-AUTOMATIC ZERO)**).
- **PEAK**: keeping the input closed the maximum weight value reached remains on display. Opening the input the current weight is displayed.
- **PLC**: closing the input no operation is performed, the input status may however be read remotely by way of the communication protocol.
- **CONTROL n**: closing the input for max one second the weight is transmitted over the serial connection according to the fast continuous transmission protocol only once (**only if CONTROL n is set in the item SERIAL**).
- **COEFF**: when the input is closed the weight is displayed based on the set coefficient (see setting of the units of measure and coefficient), otherwise the weight is displayed.
- **PRINT**: when the input is closed the data are sent for printing if in the communication protocol of either serial port the parameter **PRINT** is set. If the alibi memory is active, data storage is carried out too.

OUTPUTS AND INPUTS CONFIGURATION (LOAD program)



OUTPUTS

- OUTPUT 1: PRESET (for operation see **OPERATION SETTINGS**)
- OUTPUT 2: SET (for operation see **OPERATION SETTINGS**)
- OUTPUT 3: CYCLE END (the relay is closed to the achievement of Cycle End)
- OUTPUT 4: ALARM (the relay is closed when an alarm is present)
- OUTPUT 5: TOLERANCE (the relay is closed when the weight is out of tolerance)

INPUTS

- INPUT 1: START;
- INPUT 2: STOP;
- INPUT 3 (default = **ZERO**): It's possible to select one of the following functions:
 - **NET-GROSS** (NET/GROSS): by closing this input for no more than one second, it's making an operation of SEMI-AUTOMATIC TARE and the display will show the net weight. To display the gross weight again, hold the NET/GROSS input closed for 3 seconds.
 - **SEMI-AUTOMATIC ZERO** (SEMI-AUTOMATIC ZERO): by closing the input for no more than one second, the weight is set to zero (see section **SEMI-AUTOMATIC ZERO (WEIGHT ZERO-SETTING FOR SMALL VARIATIONS)**).
 - **APPROVAL** (APPROVAL): the instrument starts the batching only after verifying that this input is closed.

ALARM RELAY CLOSURE (LOAD program)



The ALARM relay closing can be enabled or disabled for each of the following alarms: approval contact (**CDnSP**); formula not programmed (**EMPTY**); maximum weight exceeded (**MASSFOR**); minimum weight (**LAKE**); no increase in weight (**LOAD**); no decrease in weight (**UNLOAD**); fall (**FALL**); the PC did not read the batching data (**SLAKE**).

YES: in presence of alarm, the relay is closed (default);

NO: the relay is not closed even in the presence of alarm.



Function valid **only** if **ALAR n = 0** (see section **SWITCHING OF THE ALARM RELAY ON WEIGHT**).

SEMI-AUTOMATIC TARE (NET/GROSS)



THE SEMI-AUTOMATIC TARE OPERATION IS LOST UPON INSTRUMENT POWER-OFF.

To perform a net operation (SEMI-AUTOMATIC TARE), close the NET/GROSS input or press the **TARE** key for less than 3 seconds. The instrument displays the net weight (just set to zero) and the NET symbol lights up. To display the gross weight again, keep the NET/GROSS input closed or press **TARE** for 3 seconds.

This operation can be repeated many times by the operator to allow the loading of several products.

Example:

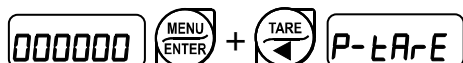
Put the box on the scale, the display shows the box weight; press **TARE**, the display shows the net weight to zero; introduce the product in the box, the display shows the product weight. This operation can be repeated several times.



While the net weight is displayed, keep  pressed to display gross weight. When the key is released the net weight will be displayed again.

The semi-automatic tare operation is not allowed if the gross weight is zero or negative or unstable or above the maximum capacity.

PRESET TARE (SUBTRACTIVE TARE DEVICE)



It is possible to manually set a preset tare value to be subtracted from the display value provided that the $P-TARE \leq \text{max capacity}$ condition is verified. In multi-interval instruments, the max permitted value is Max1 (max capacity of range 1).

By default the instrument shows the last programmed preset tare value: to apply it press  and then **ENTER**.

After setting the tare value, going back to the weight display, the display shows the net weight (subtracting the preset tare value) and the NET symbol lights up to show that a tare has been entered. To delete a preset tare and return to gross weight display, hold down **TARE** for about 3 seconds or keep the NET/GROSS input (if any) closed for the same length of time (3 seconds). The preset tare value is set to zero. The NET symbol is turned off when the gross weight is displayed once again.



While the net weight is displayed, keep  pressed to display gross weight. When the key is released the net weight will be displayed again.

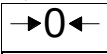


- IF A SEMI-AUTOMATIC TARE (NET) IS ENTERED, IT IS NOT POSSIBLE TO ACCESS THE ENTER PRESET TARE FUNCTION.
- IF A PRESET TARE IS ENTERED, IT'S STILL POSSIBLE TO ACCESS THE SEMI-AUTOMATIC TARE (NET) FUNCTION. THE TWO DIFFERENT TYPES OF TARE ARE ADDED.



ALL THE SEMI-AUTOMATIC TARE (NET) AND PRESET TARE FUNCTIONS WILL BE LOST WHEN THE INSTRUMENT IS TURNED OFF.

SEMI-AUTOMATIC ZERO (WEIGHT ZERO-SETTING FOR SMALL VARIATIONS)

By closing the SEMI-AUTOMATIC ZERO input, the weight is set to zero; alternatively, by pressing the  key for less than 3 seconds, the **SEt-ZERO** message is displayed for 3 seconds, by pressing **ENTER** the weight is set to zero. The zero-setting will be lost upon instrument power-off.

This function is only allowed when:

- the weight is stable; otherwise the **unStBL** alarm appears;
- the combined effect of the semi-automatic zero and zero tracking does not exceed 2% of the maximum capacity, otherwise the alarm **LT-----** appears.

PEAK (BASE program)

By keeping the PEAK input closed or by activating the function via fieldbus, the maximum weight value reached remains displayed. By opening the input or by deactivating the function via fieldbus, the current weight is displayed.



If you wish to use this input to view a sudden variation peak, set the **FILTER ON THE WEIGHT** to 0.

Peak values have no legal validity.

ANALOG OUTPUT (TLM8 Rev 1 only)



- **TYPE**: it selects the analog output type (4÷20 mA, 0÷20 mA, 0÷10 V, 0÷5 V; default: 4÷20 mA).
- **MODE**: operating mode
 - **GROSS**: the analog output varies according to gross weight
 - **NET**: the analog output varies according to net weight
 - **PLC**: remote control (see Communication protocols manual)
 - **PEAK**: the analog output transmits the PEAK weight
- **ERROR**: it sets the analog output value in error condition (see table ANALOG OUTPUT ALARM in section **ALARMS**)
- **ANAL 0**: set the weight value for which you wish to obtain the minimum analog output value.



Only set a value different from zero if you wish to limit the analog output range; for instance: for a full scale value of 10000 kg you require an 4 mA signal at 5000 kg and 20 mA at 10000 kg, in this case, instead of zero, set 5000 kg.

- **ANAL FS**: set the weight value for which you wish to obtain the maximum analog output value; it must correspond to the value set in the PLC program (default: calibration full scale). E.g.: if I am using a 4÷20 mA output and in the PLC program I wish to have 20 mA = 8000 kg, I will set the parameter to 8000.
- **CDR 0**: analog output correction to zero: if necessary adjust the analog output, allowing the PLC to indicate 0. The sign “-” can be set for the last digit on the left. E.g.: if I use a 4÷20 mA output and, with the minimum analog setting, the PLC or tester read 4.1 mA, I must set the parameter to 3.9 to obtain 4.0 on the PLC or tester.
- **CDR FS**: correction of analog output to full scale: if necessary permit modification of the analog output by allowing PLC to indicate the value set in the parameter **ANAL FS**. E.g. if I am using a 4÷20 mA output with the analog set to full scale and the PLC or tester reads 19.9 mA, I must set the parameter to 20.1 to get 20.0 on the PLC or tester.

Minimum and maximum values which can be set for zero and full scale corrections:

ANALOG OUTPUT TYPE	Minimum	Maximum
0÷10 V	-0.150	10.200
0÷5 V	-0.150	5.500
0÷20 mA	-0.200	22.000
4÷20 mA	-0.200	22.000

NOTE: the analog output may also be used in the opposite manner, i.e. the weight setting that corresponds to the analog zero (**ANAL 0**) may be greater than the weight set for the analog full scale (**ANAL FS**). The analog output will increase towards full scale as the weight decreases; the analog output will decrease as the weight increases.

For example:

$ANA\ 0 = 10000$ $ANA\ F5 = 0$ analog output 0÷10 V

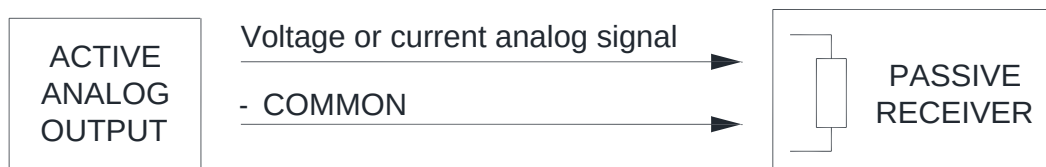
Weight = 0 kg analog output = 10 V

Weight = 5000 kg analog output = 5 V

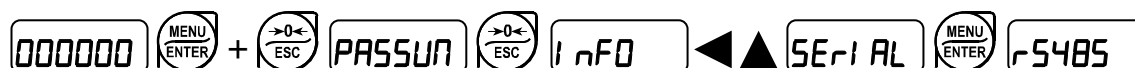
Weight = 10000 kg analog output = 0 V



All analog outputs of the instrument are ACTIVE and SINGLE ENDED type, therefore they can be connected only to PASSIVE receiver devices. The minimum load allowed for voltage outputs is 2 kohm, the maximum load allowed for current outputs is 400 ohm.



SERIAL COMMUNICATION SETTING



- **nOnE**: it disables any type of communication (default).
- **ModBUS**: MODBUS-RTU protocol; possible addresses: from 1 to 99 (see Communication protocols manual).
- **ASCII**: ASCII bidirectional protocol; possible addresses: from 1 to 99 (see Communication protocols manual).
 - **ModU60**
 - **Mod t d**
- **ContI n**: continuous weight transmission protocol (see Communication protocols manual), at the frequency set in **HEr t2** item (from 10 to 300).
 - **Mod t** (set: **PARI tY** = **nOnE**, **StOP** = 1).
 - **Mod t d** (set: **PARI tY** = **nOnE**, **StOP** = 1).
- **rI P**: continuous weight transmission protocol to remote displays; the remote display shows the net weight or gross weight, according to its settings (set: **bAUd** = 9600, **PARI tY** = **nOnE**, **StOP** = 1) (see Communication protocols manual).
- **Hd-rI P**: continuous weight transmission protocol to remote displays; the remote display shows the net weight or gross weight, according to its settings (set: **bAUd** = 9600, **PARI tY** = **nOnE**, **StOP** = 1) (see Communication protocols manual).

- **Hdrl Pn**: continuous weight transmission protocol to remote displays (set: **bAUD = 9600**, **PARITY = nOnE**, **STOP = 1**) (see Communication protocols manual).

When the remote display is set to gross weight:

- if the instrument displays the gross weight, the remote display shows the gross weight.
- if the instrument shows the net weight, the remote display shows the net weight alternated with the message **nEt**.
- **YHL**: continuous weight transmission protocol.
- **PrI nEr**: printer.

- **bAUD**: transmission speed (1200, 2400, 4800, 9600, 19200, 38400, 115200; default: 9600).
- **Addr**: instrument address (from 1 to 99; default: 1).
- **HErEz**: maximum transmission frequency (10 – 20 – 30 – 40 – 50 – 60 – 70 – 80 – 100 – 200 – 300; default: 10); to be set when the **COntI n** transmission protocol is selected.

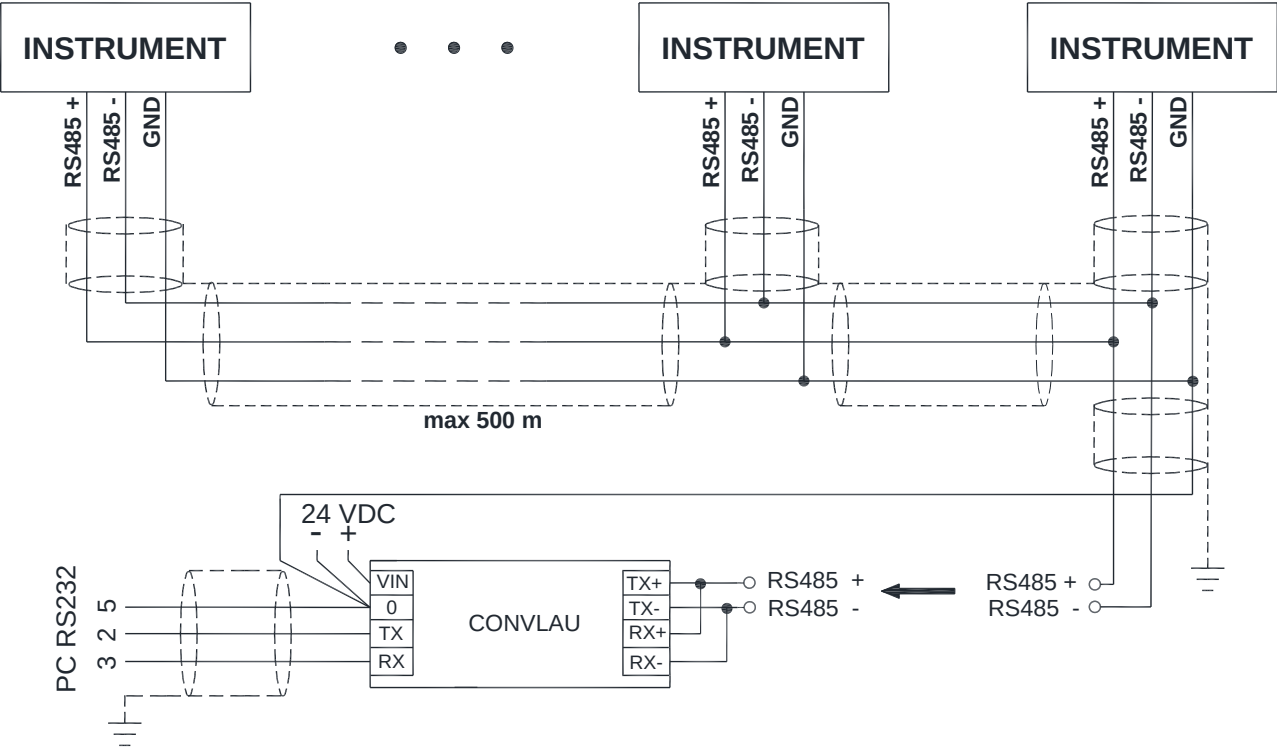
Maximum setting frequency (**HErEz**):

- 20 Hz with minimum baud rate 2400 baud.
- 40 Hz with minimum baud rate 4800 baud.
- 80 Hz with minimum baud rate 9600 baud.
- 100 Hz with minimum baud rate 19200 baud.
- 200 Hz with minimum baud rate 38400 baud.
- 300 Hz with minimum baud rate 38400 baud.
- **dELAY**: delay in milliseconds which elapses before the instrument replies (from 0 to 200 ms; default: 0).
- **PARITY**:
 - **nOnE**: no parity (default).
 - **EUEn**: even parity.
 - **Odd**: odd parity.
- **STOP**: stop bit (1 – 2; default: 1).
- **St bl t**: stability character (**YES – n0**; default: **n0**); to be set when the **COntI n** transmission protocol is selected in **NOd t** mode (see section **CONTINUOUS FAST WEIGHT TRANSMISSION PROTOCOL** in Protocols manual).
- **nCOPY**: number of copies (1 – 9; default: 1).
- **ENPEY**: number of blank lines between one printout and the next.
- **HEADEr**: printing of custom heading from PC (**YES – n0**; default: **n0**).
- **PrtNOd**: connected printer type:
 - **P 190**
 - **StAUP**
 - **StAUE**

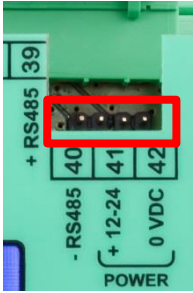


For more information about protocols and methods of communication, request the proper manual to technical assistance.

RS485 SERIAL COMMUNICATION



If the RS485 network exceeds 100 metres in length or baud-rate over 9600 are used, two terminating resistors are needed at the ends of the network: close the two jumpers indicated in the picture on the furthest instruments. Should there be different instruments or converters, refer to the specific manuals to determine whether it is necessary to connect the above-mentioned resistors.



DIRECT CONNECTION BETWEEN RS485 AND RS232 WITHOUT CONVERTER

Since a two-wire RS485 output may be used directly on the RS-232 input of a PC or remote display or printer, it is possible to implement instrument connection to an RS-232 port in the following manner:

INSTRUMENT		RS232
RS485 -	→	RXD
RS485 +	→	GND



This type of connection allows A SINGLE instrument to be used in a ONE WAY mode.

AUTOMATIC DIAGNOSTICS OF LOAD DISTRIBUTION



Only use this function in systems where load distribution can be repeated with each change of weight (for example: liquid weighing).

- **AUT (YES/NO; default: NO)**: it enables load diagnostics.
- **STATUS**: it displays the active channels status (weight, load percentage on each channel, enabled channels and/or channels in error).
- **PERI P**: it displays stored load distributions (weight, load percentage on each channel, number distribution saved).
- **DELETE**: it deletes stored load distributions; confirmation is requested (**SURE?**), press **ENTER** to proceed or press **ESC** to cancel.
- **ERR (default: 5.0)**: difference between the current and stored percentage beyond which the **Err! AG** alarm is triggered.
- **AUT 0 (YES/NO; default: NO)**: it enables diagnostics on zero.
- **ERR0 (default: 5.0)**: difference between the current and stored percentage beyond which the **Err! AG** alarm is triggered.
- **COMP 0**: it displays current load distribution on zero and the one previously stored (weight, load percentage on each channel).

Load diagnostics: the instrument, with stable weight, calculates and stores the load percentage on each channel. If under normal operation the load percentage error is higher than the value set in parameter **ERR**, the display shows the **Err! AG** alarm alternated with the weight; the alarm remains active also upon instrument power-off, press **ENTER** to cancel.

Diagnostics on zero: if with unloaded system the load percentage error is higher than the value set in parameter **ERR0**, the display shows the **Err! AG** alarm alternated with the weight; the alarm remains active also upon instrument power-off, press **ENTER** to cancel.



Diagnostics on zero is only available if it has been enabled during the installation.

Example of current load distribution display: the top of the display shows the weight on the scale; the bottom shows the load percentage on each active channel.

This screen can also be accessed directly from the weight display by pressing the **TEST** key.



TEST



- **Load distribution:**
STATUS: it displays the active channels status (weight, load percentage on each channel, enabled channels and/or channels in error). This menu can also be accessed directly from the weight display by pressing the **TEST** key.
- **Input Test:**
IN: ensure that for each open input **I** is displayed, **1** is displayed when the input is closed.
- **Output Test:**
OUT: setting **0** ensure that the corresponding output opens. Setting **1** ensure that the corresponding output closes.
- **Analog Output Option Test (TLM8 Rev 1 only):**
ANALOG: It allows the analog signal to range between the minimum and the maximum values starting from the minimum.
PA: current output test.
VOLT: voltage output test.
- **Millivolt Test:**
mv-CEL: it displays the response signal of each load cell expressed in mV with three decimals.
- **Millivolt stored at zero setting (only if AUTO = YES):**
mv-0-0: it displays the response signal of each load cell expressed in mV with three decimals.
- **Weight in E/10 test:**
E/10: it displays the current weight in tenths of **E**, for testing purposes.

EVENTS LOG



The instrument can store up to 50 events; the oldest records are overwritten.

- **dl 5 EU:** it displays the last 50 events stored, starting from the most recent one:
 - ZERO:** zero-setting from the calibration menu, press **ENTER** to display the value set to zero.
 - FtEO:** theoretical calibration, press **ENTER** to display the full scale set.
 - FREA:** real calibration, press **ENTER** to display the sample weight used.
 - INPD:** tare setting via the keypad, press **ENTER** to display the set value.
 - dl AC:** load distribution error, press **ENTER** to display the weight value that triggered the alarm, press **ENTER** again to display the difference between the load percentage and the stored value.
 - ALL:** weight alarm, press **ENTER** to display the alarm type.
 - dELr:** load distribution deletion.

EQU: equalization.

CHn: modification or deletion of the manual selection of active channels.

CHOL: overload error, press **ENTER** again to view the channels that generated the alarm.

- **DEL EU**: it deletes stored events; confirmation is requested (**SURE**), press **ENTER** to proceed or press **ESC** to cancel.
- **Print EU**: it prints all events.

INFO MENU



GENERAL: the identification data of the instrument are displayed.

- **Instn**: instrument model
- **SW CD**: software code
- **PRG**: program type
- **FU VER**: software version
- **SER n**: serial number

LEGAL: the main legally relevant parameters are displayed.

- **PRG**: approval status of the instrument
- **nDn**: active standard on the instrument
- **REFUN**: number of qualified accesses
- **FS-LEO**: system full scale
- **SENSI b**: rated sensitivity of the load cells
- **E**: scale verification division for single interval instruments
- **E 1**: scale verification division for the first range of multi-interval or multiple range instruments
- **E 2**: scale verification division for the second range of multi-interval or multiple range instruments
- **E 3**: scale verification division for the third range of multi-interval or multiple range instruments
- **MAH**: maximum capacity for single interval instruments
- **MAH 1**: maximum capacity of range 1 for multi-interval or multiple range instruments
- **MAH 2**: maximum capacity of range 2 for multi-interval or multiple range instruments
- **MAH 3**: maximum capacity of range 3 for multi-interval or multiple range instruments
- **G CAL**: gravity acceleration value in the calibration area
- **G USE**: gravity acceleration value in the place of use
- **TEMP COR**: weight correction depending on temperature
- **RELOAD**: restoring the printing/storage function
- **unit E**: valid unit of measure when the coefficient function is active
- **COEFF**: weight display change coefficient
- **ENABLE**: alibi memory status

SETPOINT PROGRAMMING (BASE program)

From the weight display, press **MENU** to access the setpoint setting.

MENU/ENTER: to enter a menu/confirm the data entry.

▲: to modify the displayed figure or menu item.

◀: to select a new figure or modify the displayed menu item.

ESC: to cancel and return to the previous menu.

000000
MENU
ENTER
SEt 1
...
SEt 5
MENU
ENTER
HYSEt 1
...
HYSEt 5
MENU
ENTER
CHSEt

- **SEt** (from 0 to full scale; default: 0): Setpoint; relay switching occurs when the weight exceed the value set in this parameter. The type of switching is settable (see section **OUTPUTS AND INPUTS CONFIGURATION (BASE program)**).
- **HYSEt** (from 0 to full scale; default: 0): Hysteresis, value to be subtracted from the setpoint to obtain contact switching for decreasing weight. For example with a setpoint at 100 and hysteresis at 10, the switching occurs at 90 for decreasing weight.
- **CHSEt** (from 0 to full scale; default: 0): gross weight value for each channel, after which the contact switches, in case of associated function **CHOL** (see section **OUTPUTS AND INPUTS CONFIGURATION (BASE program)**).

USE WITH W SERIES INSTRUMENTS (BASE program)

When the TLM8 is used in combination with a W serie instrument, the load cells are connected to the multi-channel weight transmitter, which transmits the weight to the indicator.

CONNECTION TO THE WEIGHT INDICATOR

Connection via **RS485** port.

SIGNAL	TLM8	Weight indicator
RS485 +	39	RS485 +
RS485 -	40	RS485 -
SHIELD	42	SHIELD

Weight indicator configuration

Configure, on the serial port used, the **RS485** protocol with the following parameters:

BAUD = 9600 **PARITY** = NONE **STOP** = 1

TLM8 multi-channel weight transmitter configuration

Configure, on the serial port used, the MODBUS protocol with the following parameters:

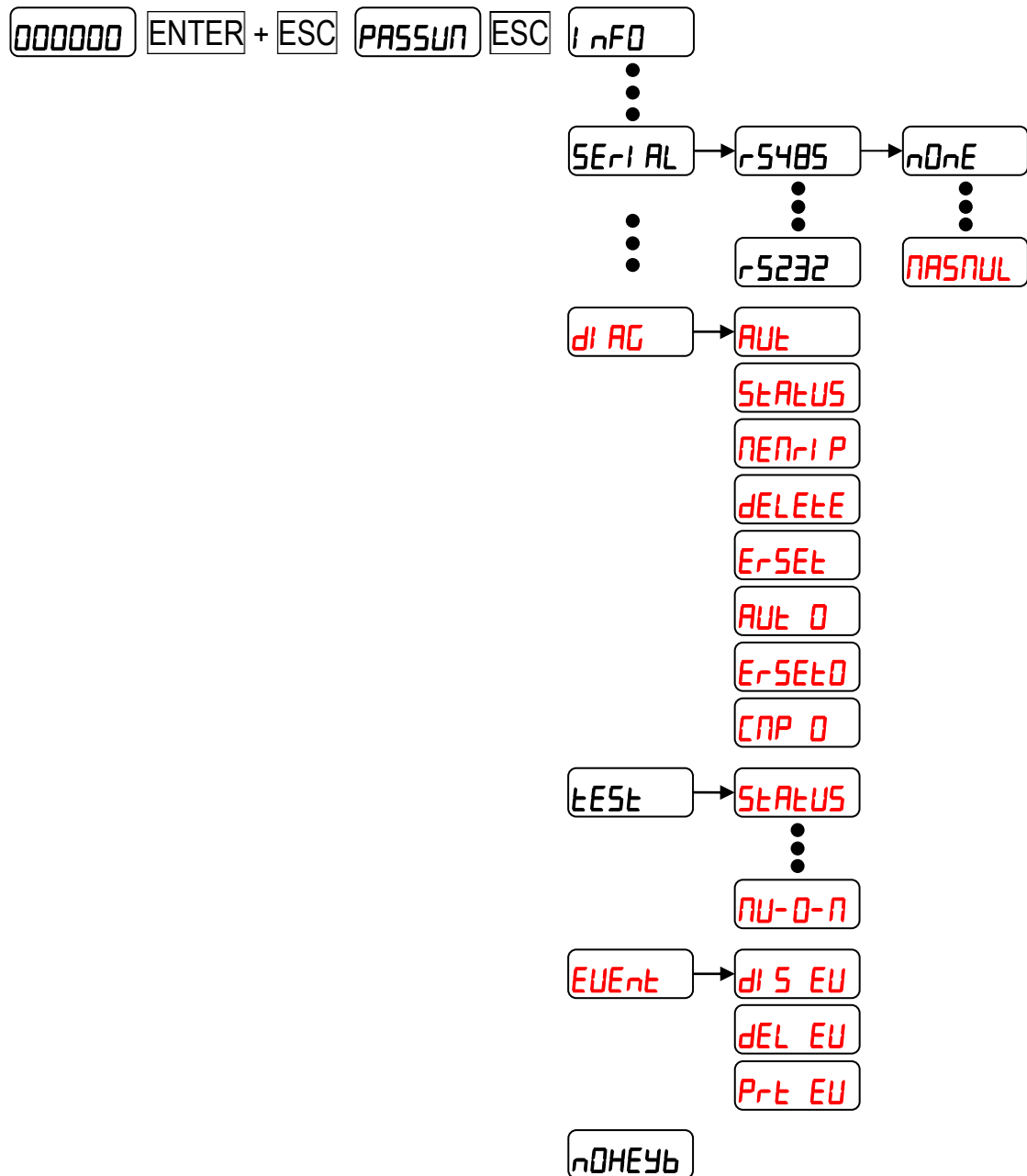
BAUD = 9600 **PARITY** = NONE **STOP** = 1

ADDITIONAL MENU MAP



WARNING: the map shows only the additional menu items that are enabled on the W series instruments connected to the TLM8.

Into menus changes are applied right after pressing the **ENTER** key (no further confirmation is required).



TLM8 REMOTE CONTROL

The following functions of TLM8 can be performed remotely through the weight indicator:

- FILTER ON THE WEIGHT
- ANTI PEAK
- AUTOMATIC DIAGNOSTICS OF LOAD DISTRIBUTION
- LOAD DIAGNOSTICS
- DIAGNOSTICS ON ZERO
- TEST
- EVENTS LOG



WARNING: when using the weight indicator to manage the multi-channel instrument, the weight indicator display replicates exactly what would be displayed on the TLM8 display.

If the configuration is performed on the TLM8, the W series instrument connected must be restarted to allow synchronization.

REMOTE KEYPAD LOCKING

`000000` `ENTER` + `ESC` `PASSUN` `ESC` `INF0` ◀ ▶ `0HEyb`

`YES`: keypad locked.

`n0`: keypad unlocked.

ADDITIONAL ALARMS



WARNING: the list shows only the additional alarms that are enabled on the W series instruments connected to the TLM8.

Er CON: TLM8 is not responding, check connections and serial ports settings.

Er rES: TLM8 responds incorrectly, check connections and serial ports settings.

ErUARt: issues with the serial communication device.

SynC: the instrument is synchronizing with TLM8, wait for the end of the operation.

ALARMS

- nOCEL:** no load cell detected, check the connections.
- ErCEL:** the load cell signal exceeds 39 mV; the conversion electronics (AD converter) is malfunctioning.
- ErCELR:** the load cell excitation is not connected or is incorrectly connected; the references are not connected or are incorrectly connected; the load cell is a 4-wire and there are no jumpers between EX- and REF- and between EX+ and REF+.
- ErCEL I:** the load cell is not connected or is incorrectly connected (the number indicates the channel on which the error is detected).
- Er DL:** the weight display exceeds 110% of the full scale.
- Er Ad:** internal instrument converter failure; check load cell connections, if necessary contact technical assistance.
- :** the weight exceeds the maximum capacity by 9 divisions.
- Er DF:** maximum displayable value exceeded (value higher than 999999 or lower than -999999).
- E-----:** weight too high: zero setting not possible.
- PAH-PU:** this message appears in the sample weight setting, in real calibration, after the eighth sample weight value has been entered.
- Error:** the value set for the parameter is beyond the permitted values; press **ESC** to quit the setting mode leaving the previous value unchanged. Examples: a number of decimals is selected for full scale which exceeds the instrument's display potential; value above the maximum setting value; the weight value set in sample weight verification does not match the detected mV increase; the analog output correction goes beyond the permitted limits.
- bLOC:** lock active on menu item, keypad or display.
- nOdl SP:** it's not possible to display properly the number because is greater than 999999 or less than -999999.
- EUI CE:** the current load cell has already been equalized; press **ENTER** to go back to the previous step and move the sample weight on the next load cell.
- LOAd:** the sample weight was not loaded or is too light.
- Erdl AG:** the load percentage error is higher than the value set in parameters **ErSEt** or **ErSEt0**; press **ENTER** to cancel the alarm.
- :** the weight is below -20e, or the power supply is insufficient.
- ALI FUL:** the alibi memory is full and will be overwritten.
- unStbL:** unstable weight: zero-setting or net weight not possible.
- nEG-0:** below zero or zero weight: net weight not possible.
- n nLEG:** weight below minimum weight (20e): not possible to print or save.
- nCHAnG:** weight has not changed: not possible to print or save.
- bUS Er:** issues with the fieldbus device.
- CHOL I:** the weight detected by the channel indicated by the number is greater than that set in the **CHSEt** parameter.
- ErrCHH:** issues with data contained in external memory, contact technical assistance.
- nOJUNP:** the jumper required for qualified access is missing (only if the active standard on the instrument is INMETRO).

Serial protocol alarms:

	<i>ErCEL</i>	<i>Er OL</i>	<i>Er Ad</i>	-----	<i>Er OF</i>	<i>t-----</i>	-----
MODE							
Bit LSB	76543210 xxxxxxxx1	76543210 xxxxx1xxx	76543210 xxxxxxxx1x	76543210 xxxxxx1xx	76543210 On gross: xxx1xxxx On net: xx1xxxxx	The response to the zero command is a 'value not valid' error (error code 3)	76543210 x1xxxxxx
Status Register MODBUS RTU							
ASCII	<u>O-F</u>	<u>O-L</u>	<u>O-F</u>	<u>O-L</u>	<u>O-F</u>	&aa#CR	<u>O-L</u>
RIP *	<u>O-F</u>	<u>O-L</u>	<u>O-F</u>	<u>O-L</u>	<u>O-F</u>	<u>O-F</u>	<u>O-L</u>
HDRIP-N	ERCEL	ER OL	ER AD	#####	ER OF	O SET	#####
CONTIN	ERCEL	ER OL	ER AD	^^^^^^	ER OF	O SET	^^^^^^
YHL	999999999	999999999	999999999	999999999	999999999	999999999	999999999

* For RIP remote displays, if the message exceeds 5 digits the display reads -----.

In case of alarm the relays open and the analog outputs are brought to the value indicated in the table (see section ANALOG OUTPUT (TLM8 Rev 1 only)):

ANALOG OUTPUT ALARM				
<i>ErrURL</i>	<i>TYPE</i>			
	0÷20 mA	4÷20 mA	0÷5 V	0÷10 V
<i>Std</i>	0 mA	3.5 mA	0 V	0 V
<i>FULL</i>	20 mA	20 mA	5 V	10 V

BATCHING (LOAD program)

INTRODUCTION TO THE OPERATION

The instrument is able to load automatically a settable amount of product on the weighing structure, driving the batching organ (including two-speed) through the PRESET and SET contacts.

The instrument has the following features:

- 1 settable formula (see section **FORMULA PROGRAMMING**);
- automatic fall calculation (see section **FALL**);
- autotare at batching start (see section **AUTOTARE**);
- tolerance error control (see section **TOLERANCE**);
- precision batching through slow function (see section **SLOW**);
- precision batching through tapping function (see section **TAPPING FUNCTION**);
- consumption storage (see section **CONSUMPTION**).

BATCHING START:

via external contact or remotely via fieldbus.

CONSUMPTION STORAGE:

the instrument, at the end of every batching, stores the consumed amount. Consumptions value can be viewed directly on instrument display or via the communication protocol when the instrument is in standby mode.

ALARM MANAGEMENT:

if an alarm occurs during the batching, the instrument shows the alarm on the display and waits for operator intervention to abort or continue the batching.

BATCHING SEQUENCE

The batching process consists of the following steps:

- batching sequence programming;
- batching start;
- batching execution;
- waiting phase;
- cycle end phase.

BATCHING SEQUENCE PROGRAMMING

In case of remote control of the instrument, select the number of cycles to run.

BATCHING START

Start the batching by closing the START external contact for at least 0.5 s.



If at the end of the batching the START contact is closed, the same batching sequence is repeated.

The instrument will ensure that:

- the formula has been programmed, otherwise it will display the alarm **ENPTY**;
- the amount set in the formula does not exceed the set maximum weight (**MASS**), otherwise the message **MASSFL** is displayed;
- IN3 input (if it is set as Approval) is closed, otherwise the message **CONSP** will be displayed; close the IN3 input to start the batching.
- the gross weight is lower than the minimum weight (**MIN**), otherwise the alarm **LTARE P** is displayed (however it is possible to force batching start by pressing **ENTER**).
- Only if **RLARE** is set: the following conditions are met to perform the autotare:
 - the gross weight is greater than zero, otherwise the alarm **NEG-0** is displayed;
 - the grow weight is stable, otherwise the alarm **UNSTBL** is displayed;
- Only if **RZER0** is set: the following conditions are met to perform the autozero:
 - the gross weight is stable, otherwise the alarm **UNSTBL** is displayed;
 - the total zero-setting does not exceed 2% of the maximum capacity, otherwise the alarm **L-----** is displayed;
- The amount programmed in the formula must exceed 20 divisions; otherwise the message **MINLEG** will be displayed.
- only if **LOLAL = YES**:
 - if the consumed amount exceeds 999000, the **LOLAL** message appears for a second;
 - if the consumed amount exceeds 999999, it is automatically set to zero.

Once these conditions have been verified, the instrument proceed to the batching execution phase.

BATCHING EXECUTION

The batching consists of an initial FAST phase and a subsequent SLOW phase.

During the entire batching execution, the instrument:

- monitors the weight of the product's batched quantity;
- commands the PRESET and SET contacts according to the **P55** parameter setting (see section **OPERATION SETTINGS**);
- if the **ENLDRd** time is set and the product is not loaded by at least 20 divisions within this interval of time, activates the **LDRdP** alarm.

The FAST phase ends when one of the following conditions occurs:

- only if $SLDU$ is set: the batched quantity has reached the set value minus the slow value;
- the batched quantity has reached the preset value set in formula.

During the SLOW phase:

- only if $ENCOP$ is set: the instrument does not verify the reaching of set value until the **NO COMPARISON TIME** has elapsed;
- if a tapping value has been set, the SET contact performs opening and closing cycles according to the times set by $SLDUDn$ and $SLDUOF$ parameters.

When the batched quantity reaches the set value configured in the formula minus any fall value, the SLOW phase ends and the SET contact opens, interrupting the product flow.

WAITING PHASE

After opening the SET, the instrument shows the weight preceded by the letter **A** and waits:

- The waiting time has elapsed ($ENEAR$);
- The START input has been closed or the **ENTER** key has been pressed or confirmed via fieldbus;
- The weight is stable.

If the tolerance (EDL) is set, three conditions can occur:

- the batched quantity is less than the set quantity minus the tolerance value; the SLOW phase is performed again to improve the batching accuracy. If the error condition persists, the EDL alarm is activated;
- the batched quantity is greater than the set quantity plus the tolerance value; the EDL alarm is enabled;
- the batched quantity falls within the tolerance limits.

To cancel the EDL alarm and proceed, press **ENTER** or confirm via fieldbus.

When all the set conditions are met, the waiting phase ends and the batching cycle continues.

CYCLE END PHASE

The system enters the cycle end phase:

- the CYCLE END contact is closed;
- only if $EDEL = YES$: consumption is stored;
- From the beginning of the product extraction, if the $ENULO$ time is set and the product is not extracted for at least 20 divisions within this interval of time, the $UNLOAd$ alarm is activated.

The instrument ends the batching, and opens the CYCLE END contact, only after verifying that:

- the weight is lower than the minimum weight ($PI n$);
- the safe emptying time has elapsed ($ESI E$).

Only if $SLAVE = YES$: the instrument waits for remote data reading, before being available for a new batching.

If several batching cycles have been set, the instrument starts a new cycle.

OPERATION SETTINGS



P55 (default: **2**): select the operating mode of SET and PRESET contacts.

- **P55 = 1**: at the batching start, only the PRESET contact is closed; once reached the preset set value, the related contact is opened and the SET contact is closed; reached the final value of set, the related contact is opened;
- **P55 = 2**: at the batching start, the SET and PRESET contacts are closed simultaneously; once reached the preset value, the related contact is opened (beginning of the slow phase); once reached the set value, also the related contact is opened. For single-speed batching, program **P55 = 2** and use only the SET contact;
- **P55 = 3**: at the batching start, only the PRESET contact is closed; once reached the preset value, also the SET contact is closed; once reached the final value of set, both are reopened.

EndnEt (default: **n0**): select the display type of net / gross weight during the CYCLE END phase (unloading of the scale at the end of batching).

- **YES**: during the cycle end, the net weight is displayed. After the CYCLE END opening, the gross weight is displayed;
- **n0**: during the cycle end, the gross weight is displayed.

DISPLAYING DURING BATCHING

During the batching the display shows the weight preceded by the letter **b**.

By pressing  the formula number is displayed, by pressing it again the number of running cycle is displayed.

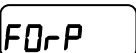
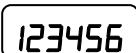
BATCHING STOP

Open the START contact and close the STOP contact to stop the batching. If the START contact is closed, the **StArT** alarm is displayed.

Alternatively, stop the batching via fieldbus.

CONSUMPTION

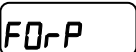
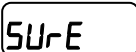
If in constants consumption is enabled (**EDt = YES**), the quantities batched are stored.

Selecting **EDtAL** the total quantity consumed (divided by 100) is displayed.

The quantities will be printed out in full.

CONSUMPTION DELETION

To delete consumption confirm **CAnCtD**, the **SUrE** deletion confirmation is requested; confirm again with  or press  to cancel the command.

PROGRAMMING OF BATCHING CONSTANTS

From weight display press **MENU**, then press  several times until **COnt** is displayed and confirm.

- MENU/ENTER**: to enter a menu/confirm the data entry.
: to modify the displayed figure or menu item.
: to select a new figure or modify the displayed menu item.
ESC: to cancel and return to the previous menu.

MINIMUM WEIGHT

Min (from 0 to full scale; default: 10): minimum weight, value at which the scale is considered empty. Batching start is only allowed if the weight is lower than this value, during the unloading phase the CYCLE END contact will be opened when the weight reaches this value and after the safe emptying time is over.

MAXIMUM WEIGHT

MASS (from 0 to full scale; default: 0): settable and displayable maximum weight. If the displayed weight exceeds the maximum weight by 9 divisions, the **-----** message is displayed; if in the formula programming the weight value set is greater than this value, the **Error** message is displayed and the value will not be stored.

By setting 0, the function is disabled.

SAFE EMPTYING TIME

ESI (from 0.0 to 999.9 seconds; default 5.0): time that is necessary for a perfect emptying of the scale. The instrument waits for this time during the unloading phase (CYCLE END closed), after reaching the minimum weight and before opening the CYCLE END contact to obtain a perfect emptying of the scale.

WAITING TIME

ENPEAL (from 0.0 to 999.9 seconds; default 5.0): time elapsing between the SET end batching and the CYCLE END closing to allow the weight to get steady. This waiting time is only required if the automatic fall is programmed and/or consumption is enabled and/or printing is enabled and/or a tolerance value has been programmed and/or confirmation from PC is enabled.

NO COMPARISON TIME

ENCOMP (from 0.0 to 999.9 seconds; default: 0.0): this is the instrument waiting time during batching, after the opening of PRESET, before comparing the weight with the programmed SET value.

NO PRODUCT LOAD TIME

ENLOAD (from 0.0 to 999.9 seconds; default: 0.0): this parameter allows the product control during batching. if there is no product load, the instrument waits for a set duration of time before activating the alarm **LOAD**.

NO PRODUCT UNLOAD TIME

ENUNLOD (from 0.0 to 999.9 seconds; default: 0.0): this parameter allows the product control during the unloading phase (cycle end). If there is no product extraction, the instrument waits for a set duration of time before activating the alarm **UNLOAD**.

WARNING: The control is only actuated after unloading of the product has started (the weight must decrease by at least 10 divisions).

FALL

By FALL it means the correction of the product amount in fall after the batching STOP. This amount is in addition to the product already batched causing inaccuracy. The instrument is able to anticipate the batching STOP, to reduce this uncertainty, with two possible ways:

- AUTOMATIC fall: the instrument automatically calculates the fall;
- MANUAL fall: the instrument applies the fall set by the operator;

WARNING: Setting a value of **WAITING TIME** (**EL PERL**) such that the weight is stable at the end of the batching, otherwise the update of the AUTOMATIC fall is not correct.

NOdFAL

NOdFAL (from 1 to 99; default: 0): it's possible to select the automatic or the manual fall.

- **NOdFAL** = 0: MANUAL fall;
- **NOdFAL** different from 0: AUTOMATIC fall (the set value shows how many batchings the fall value is updated).

Note: In the batching cycles in which the fall value is not updated, the **WAITING TIME** is not applied, unless there are no other functions that require the **WAITING TIME** (eg prints...). So, by setting a high value, it reduces the duration of batchings.

Example: If **NOdFAL** = 3 the AUTOMATIC fall is calculated every three batchings cycles.

FALdl μ (only if **NOdFAL** $\neq$ 0)

FALdl μ (from 0 to full scale; default: 0): this parameter indicates the limit within which the automatic fall is updated according to the parameter **NOdFAL**. By setting a low value is obtained more accurately but the length of the batchings could increase, because, if necessary, the AUTOMATIC fall is updated every cycle regardless of **NOdFAL**. By setting 0, the function is disabled.

Example: by setting **NOdFAL** = 3, **FALdl** μ = 5 and the weight equal to 100, the instrument updates the fall every batching, until the batched weight falls within the 95 to 105 range, and then it goes back to update the AUTOMATIC fall every 3 batchings.

FALL

FALL (from 0 to maximum weight; default: 0): in this parameter it is possible to set the fall value (only if **NOdFAL** = 0) or to display and/or to modify the fall value calculated automatically by the instrument (only if **NOdFAL** $\neq$ 0).

TOLERANCE

TOL (from 0 to maximum weight; default: 0): parameter that defines how much the batched weight value can deviate from the one set in formula. If the batched weight is higher or lower than the amount to be batched, for a value greater than the tolerance, the instrument closes the TOLERANCE contact and shows the **TOL** alarm. Press **ENTER** or confirm via fieldbus to continue the batching; the TOLERANCE contact remains closed until the weight reaches the minimum weight value and the safe emptying time has elapsed. By setting 0, the function is disabled.

Example: if a SET = 1000 value is set and a TOLERANCE = 100 value is set, the batched weight must not be lower than 900 or higher than 1100 to allow the instrument to continue the batching process.

SLOW

SLOW (from 0 to maximum weight; default: 0): replaces the preset value. When the weight has reached the SET value minus the value set in this parameter, the slow batching phase starts by the PRESET contact. If the set value is greater than the weight to batch, the batching will be in slow phase. By setting 0, the function is disabled.

Example: if SET = 100 and SLOW = 15, the slow phase begins when the weight reaches 85.

TAPPING FUNCTION

In the event that the batching instrument is not equipped with the speed “slow” it is possible to use this function to slow down the product batching in the final phase (through opening and closing cycles of the SET contact) and to increase the accuracy. Set in **SLOW** parameter the product amount you want to batch with the tapping function enabled.

SLOW ON

SLOWON (from 0 to 9.9; default: 0): time in which the SET relay remains closed during the SLOW phase. By setting 0, the function is disabled.

SLOW OFF

SLOWOFF (from 0 to 9.9; default: 0): time in which the SET relay remains open during the SLOW phase. By setting 0, the function is disabled.

AUTOTARE

AUTOTARE (from 0 to 999; default: 0): autotare enabling (automatic tare at batching start); the autotare will be updated every as many cycles of a single batching sequence as are set in this parameter. By setting 0, the function is disabled.

This operation is possible only if:

- The gross weight is greater than zero, otherwise the alarm **NEG-0** is displayed;
- The gross weight is stable, otherwise the alarm **UnStBL** is displayed;
- the gross weight is lower than the minimum weight (**Min**), otherwise the **LowEP** alarm message is displayed.

AUTOZERO

AUTOZERO (from 0 to 999; default: 0): autozero enabling (automatic zero at the batching start); the autozero procedure will be performed every as many batching operations as are set in this parameter. By setting 0, the function is disabled.

Use this function to reset the "fouling", i.e. product residues left after a scale unloading.

This operation will be performed only if:

- The gross weight is stable, otherwise the alarm **UnStBL** is displayed.
- The sum of all the zero-settings does not exceed 2% of the maximum capacity, otherwise the alarm **Exceed** is displayed.

AUTOTARE OR AUTOZERO DELAY

Delay (only if **AUTOTARE** or **AUTOZERO** $\neq$ 0, from 0 to 99.9; default: 0): the instrument waits for this time, after the starting of the formula, before performing the autotare/autozero and starting the batching.

CONSUMPTION

Consumption (default **Off**): enabling consumption memory (total batched quantity).

- **On**: consumption enabled
- **Off**: consumption disabled.

PRINT AT CYCLE END

Print (default **no**): function enabling to print batching data at cycle end.

- **YES**: print enabled
- **no**: print disabled

WAITING CONFIRMATION FROM PC (SLAVE)

SLAVE (default: **no**): waiting for confirmation of record batching data from PC.

- **YES**: check enabled; the instrument waits for the PC to read the batching data, before starting another batching. In case of recording failure, the instrument displays an alarm **SLAVE**.
- **no**: check disabled; at cycle end, the instrument will not wait for data recording on PC before performing another batching.

SWITCHING OF THE ALARM RELAY ON WEIGHT

ALARM (from 0 to full scale; default: 0): by setting a value different from zero, the ALARM contact does not switch over in case of alarm, but behaves as a SETPOINT contact; the relay closes when the weight reaches the value set in this parameter.

SWITCHING OF THE TOLERANCE RELAY ON WEIGHT

TOLER (from 0 to full scale; default: 0): by setting a value different from zero, the TOLERANCE contact does not switch on the basis of tolerance, but behaves as a SETPOINT contact; the relay closes when the weight reaches the value set in this parameter.

FORMULA PROGRAMMING

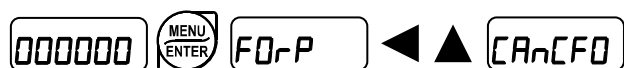
Confirm with **ENTER** the **FORP** prompt and set **PRESET** and **SET** of the batching formula.



PRESET is NOT displayed if a **SLOW** value has been set in the constants (**SLOW**).

In case of attempt in **PRESET** or **SET** to set a value lower than 20 divisions or higher than the maximum weight (**MASS**) set in the batching constants, the message **ERROR** is displayed.

DELETING FORMULA



Confirm with **ENTER** the **CANCFD** prompt, to delete the formula you will be prompted to confirm (**SURE**), to delete confirm again with **ENTER** otherwise press **ESC** to cancel the command.

ADDITIONAL ALARMS RELATING TO BATCHING

- HAEP:** it is displayed if, at batching start, the weight on the scale is higher than the minimum set in constants (m_n). Press **ESC** to return to the weight displaying, press **ENTER** to cancel the alarm and continue with the batching. If the weight comes back below to the minimum set, the batching starts.
- CONS?** it is displayed when at the batching start the APPROVAL input is open (if the input 3 is set as approval; In 3 = **CONS**). Close the input to enable batching or cancel it by pressing **ESC**.
- EDL:** it is displayed if at batching end the weight is different from the value set in formula by a value higher than tolerance. Press **ENTER** or confirm via fieldbus to cancel the alarm and continue with the batching.
- FALL:** it is displayed if at batching start the fall value is higher than the product quantity to be batched. Press **ESC** to cancel the alarm and the batching.
- EMPTY:** it is displayed if, at batching start, the formula recalled for running is not programmed. Press the key **ESC** to quit.
- NASFOR:** it is displayed if, at batching start, the formula recalled for running exceeds the maximum weight. Press the key **ESC** to quit.
- LOAD:** during the batching it indicates that the product is not loaded. It is automatically cancelled if the product increases. Otherwise stop the batching (see section **BATCHING STOP**).
- UNLOAD:** during the unload (Cycle End contact closed) it indicates that the product is not extracted. It is automatically cancelled if the product decreases. Otherwise stop the batching (see section **BATCHING STOP**).
- START?** it is displayed for 3 seconds if you try to stop the batching (by closing the STOP input) when the START input is closed. Open the START input to cancel the alarm.
- PASER:** batching is cancelled. Press **ESC** to quit. If the alarm persists, contact technical assistance.
- WEI G:** it is displayed when there is a weight alarm and it cancels the current batching. Press **ESC** to go back to the weight display and check the alarm.
- Error:** the value set for the parameter is beyond the permitted values; press **ESC** to quit the setting mode leaving the previous value unchanged. Examples: a number of decimals is selected for full scale which exceeds the instrument's display potential; value above the maximum setting value; the weight value set in sample weight verification does not match the detected mV increase; the analog output correction goes beyond the permitted limits.
- COND:** waiting for START closure or **ENTER** key pressure to continue the batching.
- SLAE:** PC has not read batching data.

PRINTING EXAMPLES

In case of BASE program: if the printer has been set (see section **SERIAL COMMUNICATION SETTING**), from the weight display press the **PRINT** key:

BASIC PRINTOUT (ALIBI DISABLED):

```
-----
TLM8   BASE      Addr:01

G              878 kg
N              589 kg
T              289 kg
```

BASIC PRINTOUT (ALIBI ENABLED):

```
-----
TLM8   BASE      Addr:01

ID:                      6
G              878 kg
N              589 kg
T              289 kg
```

BASIC PRINTOUT (PEAK ENABLED):

```
-----
TLM8   BASE      Addr:01

G              1204 kg
N              831 kg
T              373 kg
P              2103 kg
```

PRINTOUT WITH **COEFF** ENABLED:

```
-----
TLM8   BASE      Addr:01

UNIT    kg |      bar
G       1195 |      1792
N       1195 |      1792
T         0 |         0
```

ALIBI MEMORY DATA RECALL:

```
-----
TLM8   BASE      Addr:01
ALIBI DATA RECALL

ID:                      125
N              1063 kg
T              500 kg
```

LOAD DISTRIBUTION PRINT

Current distribution: press the **PRINT** key from the *STATUS* menu.

Current and stored distribution: from the *COMP* and *MEMRIP* menus, keep pressed the **PRINT** key for more than 3 seconds while the weight is displayed.

CURRENT DISTRIBUTION

```
-----
TLM8   BASE      Addr:01
CURRENT (STATUS)
GROSS   2014 kg
CH1:    23.5 %
CH2:    24.1 %
CH3:    15.5 %
CH4:    16.7 %
CH5:     9.0 %
CH6:    10.2 %
CH7:      ERR      (load cell connected but in error)
CH8:      OFF      (load cell not connected)
```

CURRENT AND STORED DISTRIBUTION

```
-----
TLM8   BASE      Addr:01
STORED (MEMRIP)
GROSS   2014 kg
CH1:    23.5 %
CH2:    24.1 %
CH3:    15.5 %
CH4:    16.7 %
CH5:     9.0 %
CH6:    10.2 %
CH7:      ERR      (load cell connected but in error)
CH8:      OFF      (load cell not connected)
GROSS   OLD      2050 kg
CH1:    25.5 %
CH2:    22.1 %
CH3:    16.5 %
CH4:    16.7 %
CH5:     9.0 %
CH6:    10.2 %
CH7:      ERR      (load cell connected but in error)
CH8:      OFF      (load cell not connected)
```

In case of LOAD program: if the printer has been set (see section **SERIAL COMMUNICATION SETTING**), from the weight display press the **PRINT** key:

- **WEIGHT**: prints the displayed weight.
- **BATCH**: prints the batching constants.
- **FORM**: prints one or all of the formulas; press **ENTER** to display **FORM I**: set the formula number to be printed or 00 to print them all.
- **LEGAL**: prints consumption (only if **LEGAL** = **YES**, see **CONSUMPTION**).

If the alibi is available, for each valid weight an ID line will be printed.

BATCHING PRINTOUT

Batching in gross weight ($A_{LE}A_{RE} = 0, A_{ZE}D = 0$)

```

:::
TLM8      LOAD      Addr:01
FORMULA:                                01
CYCLE:                                         1

```

ID:	1
G	1060 kg
SET	1000

Gross weight batching with zero-setting ($A_{LE}A_{RE} = 0$, $A_{2E}A_{RD} = 1$)

```

:::
TLM8      LOAD      Addr:01
FORMULA:                                01
CYCLE:                                         1

```

```
ID: 2
G 1047 kg
SET 1000
```

Batching in net weight ($A_{LE} = 1, A_{2E} = 0$)

```

:::
TLM8      LOAD      Addr:01
FORMULA:           01
CYCLE:           2

```

ID:	3
N	1070 kg
SET	1000
T	41 kg

CONSTANTS PRINTOUT

```
.....  
TLM8    LOAD      Addr:01  
        CONSTANTS  
MIN                        10  
MASS                        0  
TIME SIC                   5.0  
TIME WAIT                  0.0  
TIME NCOMP                 0.0  
TIME NLOAD                 0.0  
TIME NUNLOAD               0.0  
FALL                      MANUAL  
SLOW                       0  
TIME SLOWON                0.0  
TIME SLOWOF                0.0  
PSS                        2  
AUTOZERO                  NO  
AUTOTARE                  NO  
UNLOAD                    GROSS  
TOTAL                     NO  
PRINT                     NO  
SLAVE                     NO  
RELAY ALARM                0  
RELAY TOL.                0  
  
FOR.      FALL      TOLER  
01        0        100
```

FORMULA PRINTOUT

Standard formula printout (*SLOW* = 0)

```
.....  
TLM8    LOAD      Addr:01  
FORMULA:                01  
SET                        2000 kg  
PSET                      200 kg
```

Formula with slow active printout (*SLOW* = 100)

```
.....  
TLM8    LOAD      Addr:01  
FORMULA:                01  
SLOW                        100 kg  
SET                        2000 kg
```

CONSUMPTION PRINTOUT

```
.....  
TLM8    LOAD      Addr:01  
        CONSUMPTION  
TOTAL                        1148 kg
```

INFORMATION ON TYPE-APPROVED INSTRUMENTS IN THE EUROPEAN UNION

Instruments first assessed by the manufacturer: they must bear an “M” sticker and can be immediately commissioned in accordance with the Directive 2014/31/EU, art.1

Instruments assessed in two stages: they do not bear an “M” sticker. The first assessment phase has been completed by the manufacturer and includes all the tests listed by EN45501:2015, 8.2.2. The second assessment phase must be completed by the authorities in charge in the Country in which the weighing scale is installed; contact your dealer in case of any doubt as to the procedure to follow.

Metrical users will be responsible for observing the time schedule relating to the required periodical assessments after the first assessment (if provided for by law in the Country of installation of the scale).

DECLARATION OF CONFORMITY - EU

LAUMAS

Innovation in Weighing

SISTEMI DI PESATURA INDUSTRIALE - CELLE DI CARICO - BILANCE

LAUMAS Elettronica S.r.l.
Via I Maggio 6 - 43022 Montechiarugolo (PR) Italy
C.F. - P.IVA IT01661140341

Tel. (+39) 0521 683124
Fax (+39) 0521 681091

Email: laumas@laumas.it
Web: <http://www.laumas.com>

Fabbricante metrico Prot. N. 7340 Parma - R.E.A. PR N. 169833 - Reg. Imprese PR N.19393 - Registro Nazionale Pile
N. IT09060P00000982 - Registro A.E.E. N. IT08020000002494 - N. Mecc. PR 008385 - Cap. Sociale € 100.000 int. vers.

SISTEMA QUALITÀ CERTIFICATO UNI EN ISO 9001 - SISTEMA GESTIONE AMBIENTALE ISO 14001 - MODULO D: GARANZIA DELLA QUALITÀ DEL PROCESSO DI PRODUZIONE

I	Dichiarazione di conformità	Dichiariamo che il prodotto al quale la presente dichiarazione si riferisce è conforme alle norme di seguito citate.
GB	Declaration of conformity	We hereby declare that the product to which this declaration refers conforms with the following standards.
E	Declaración de conformidad	Manifestamos en la presente que el producto al que se refiere esta declaración está de acuerdo con las siguientes normas
D	Konformitäts-erklärung	Wir erklären hiermit, dass das Produkt, auf das sich diese Erklärung bezieht, mit den nachstehenden Normen übereinstimmt.
F	Déclaration de conformité	Nous déclarons avec cela responsabilité que le produit, auquel se rapporte la présente déclaration, est conforme aux normes citées ci-après.
CZ	Prohlášení o shode	Tímto prohlašujeme, že výrobek, kterého se toto prohlášení týká, je v souladu s níže uvedenými normami.
NL	Conformiteit-verklaring	Wij verklaren hiermede dat het product, waarop deze verklaring betrekking heeft, met de hierna vermelde normen overeenstemt.
P	Declaração de conformidade	Declaramos por meio da presente que o produto no qual se refere esta declaração, corresponde às normas seguintes.
PL	Deklaracja zgodności	Niniejszym oświadczamy, że produkt, którego niniejsze oświadczenie dotyczy, jest zgodny z poniższymi normami.
RUS	Заявление о соответствии	Мы заявляем, что продукт, к которому относится данная декларация, соответствует перечисленным ниже нормам.

Models: TLM8

Mark Applied	EU Directive	Standards
CE	2014/35/EU Low Voltage Directive	<i>Not Applicable (N/A)</i> for VDC type EN 61010-1:2010+A1:2019 for 230/115 VAC type
CE	2014/30/EU EMC Directive	EN 55011:2016+A1+A11:2020 EN 61000-6-2:2019 EN 61000-6-4:2019 EN 61000-4-2:2009 EN 61000-4-3:2006+A2:2010 EN 61000-4-4:2012 EN 61000-4-5:2014+A1:2017 EN 61000-4-6:2014
CEM (only if "M" mark is applied)	2014/31/EU NAWI Directive	EN 45501:2015 OIML R76-1:2006

Montechiarugolo (PR), 15/01/2026

LAUMAS Elettronica s.r.l.
M. Consonni
(Legal Representative)

Consonni Massimo

DECLARATION OF CONFORMITY -UKCA



SISTEMI DI PESATURA INDUSTRIALE - CELLE DI CARICO - BILANCE

LAUMAS Elettronica S.r.l.
Via I Maggio 6 - 43022 Montechiarugolo (PR) Italy
C.F. - P.IVA IT01661140341

Tel. (+39) 0521 683124
Fax (+39) 0521 681091

Email: laumas@laumas.it
Web: <http://www.laumas.com>

Fabbricante metrico Prot. N. 7340 Parma - R.E.A. PR N. 169833 - Reg. Imprese PR N.19393 - Registro Nazionale Pile N. IT09060P00000982 - Registro A.E.E. N. IT08020000002494 - N. Mecc. PR 008385 - Cap. Sociale € 100.000 int. vers.

SISTEMA QUALITÀ CERTIFICATO UNI EN ISO 9001 - SISTEMA GESTIONE AMBIENTALE ISO 14001 - MODULO D: GARANZIA DELLA QUALITÀ DEL PROCESSO DI PRODUZIONE

I	Dichiarazione di conformità	Dichiariamo che il prodotto al quale la presente dichiarazione si riferisce è conforme alle norme di seguito citate.
GB	Declaration of conformity	We hereby declare that the product to which this declaration refers conforms with the following standards.
E	Declaración de conformidad	Manifestamos en la presente que el producto al que se refiere esta declaración está de acuerdo con las siguientes normas
D	Konformitäts-erklärung	Wir erklären hiermit, dass das Produkt, auf das sich diese Erklärung bezieht, mit den nachstehenden Normen übereinstimmt.
F	Déclaration de conformité	Nous déclarons avec cela responsabilité que le produit, auquel se rapporte la présente déclaration, est conforme aux normes citées ci-après.
CZ	Prohlášení o shode	Tímto prohlašujeme, že výrobek, kterého se toto prohlášení týká, je v souladu s níže uvedenými normami.
NL	Conformiteit-verklaring	Wij verklaren hiermede dat het product, waarop deze verklaring betrekking heeft, met de hierna vermelde normen overeenstemt.
P	Declaração de conformidade	Declaramos por meio da presente que o produto no qual se refere esta declaração, corresponde às normas seguintes.
PL	Deklaracja zgodności	Niniejszym oświadczamy, że produkt, którego niniejsze oświadczenie dotyczy, jest zgodny z poniższymi normami.
RUS	Заявление о соответствии	Мы заявляем, что продукт, к которому относится данная декларация, соответствует перечисленным ниже нормам.

Models: TLM8

Mark Applied	UK legislation	Standards
UK CA	Electrical Equipment (Safety) Regulations 2016	<i>Not Applicable (N/A)</i> for VDC type BS EN 61010-1:2010+A1:2019 for 230/115 VAC type
UK CA	Electromagnetic Compatibility Regulations 2016	BS EN 55011:2016+A1+A11:2020 BS EN 61000-6-2:2019 BS EN 61000-6-4:2019 BS EN 61000-4-2:2009 BS EN 61000-4-3:2006+A2:2010 BS EN 61000-4-4:2012 BS EN 61000-4-5:2014+A1:2017 BS EN 61000-4-6:2014
UK CA M (only if "M" mark is applied)	Non-automatic Weighing Instruments Regulations 2016	BS EN 45501:2015

Montechiarugolo (PR), 15/01/2026

LAUMAS Elettronica s.r.l.
M. Consonni
(Legal Representative)

On our website www.laumas.com there are videos on the guidelines for correct installation of weighing systems and video tutorials on configuring our transmitters and weight indicators.

All Laumas product manuals are available online. You can download the manuals in PDF format from www.laumas.com by consulting the Products section or the Download Area. Registration is required.

Think about the environment before you print!

CERTIFICATION OF THE ENVIRONMENTAL MANAGEMENT SYSTEM
in accordance with UNI EN ISO 14001.

Laumas contributes to environmental protection by saving on paper consumption.